The phenomeno-logical robot

Jean-François Cloutier
Research Fellow
Active Inference Institute

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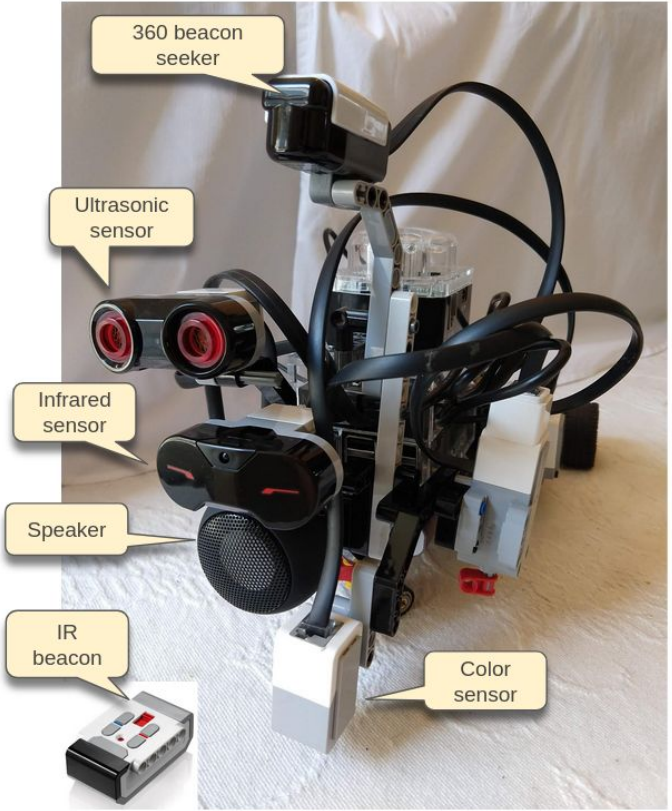
What does it take, at a minimum, for an autonomous robot to learn to survive in a world it knows initially nothing about?

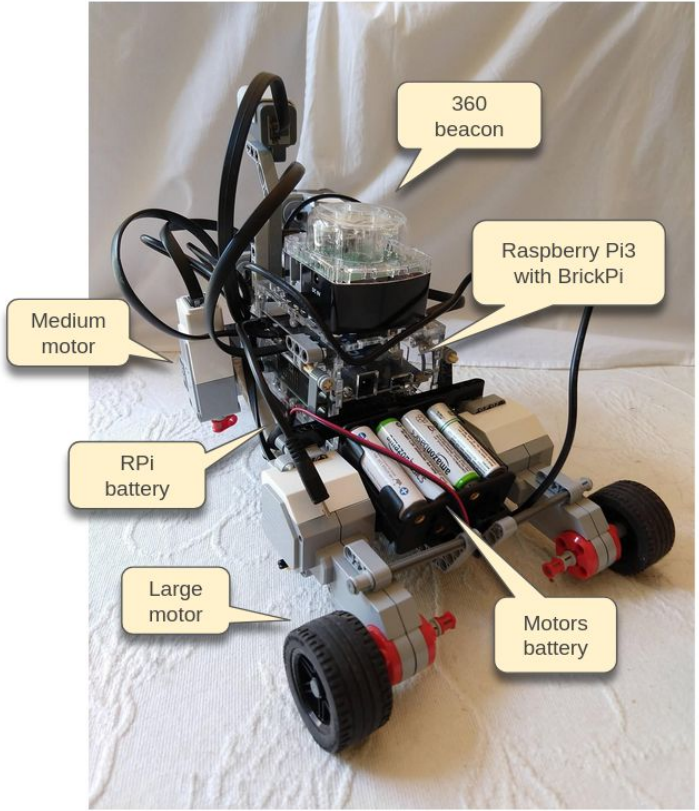
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I am programming a mortal robot to have felt experiences, make sense of them, and grow the agency it needs to survive

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The robot is a rover with Lego EV3 sensors and motors





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What this project is not about

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This is **not** an *input-output simulation* of *human-level* cognition (what LLMs do)

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This is **not** an attempt at a *general model* of cognition

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This is **not** a robot *solving* Free Energy Principle equations

But one that *complies* with FEP principles

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This is **not** a robot as an extension of its programmer

The robot's agency is grounded in its own experiences

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What this project is about

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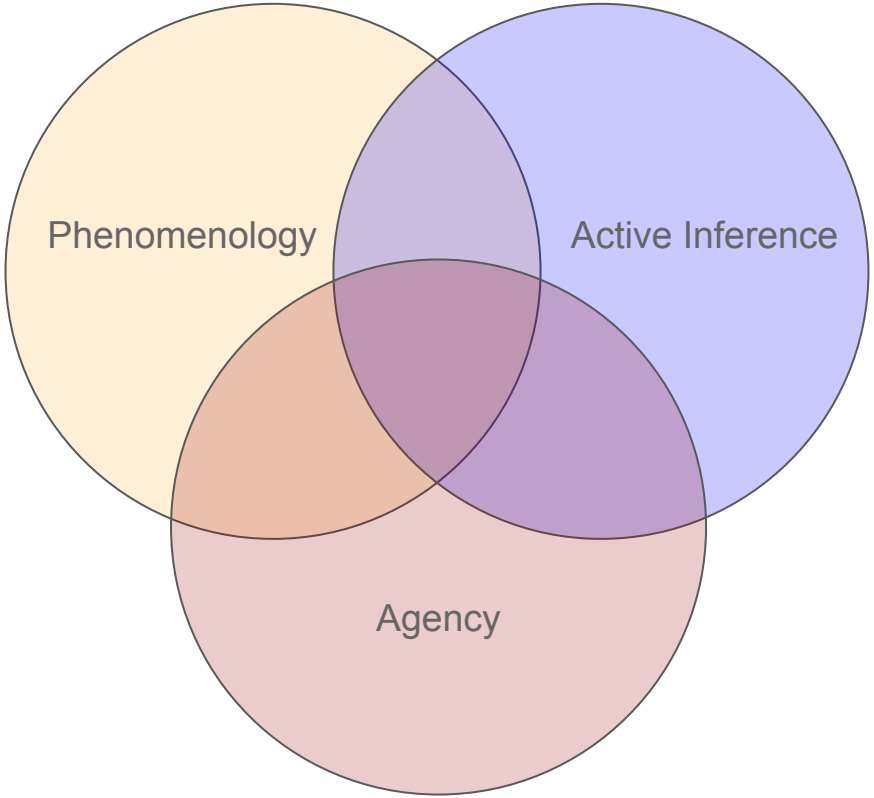
I am *building* an artificial agent

I want to *achieve basal* cognition

Experiences are to be *intrinsically* meaningful to the robot

FEP is applied as an *analytical* framework

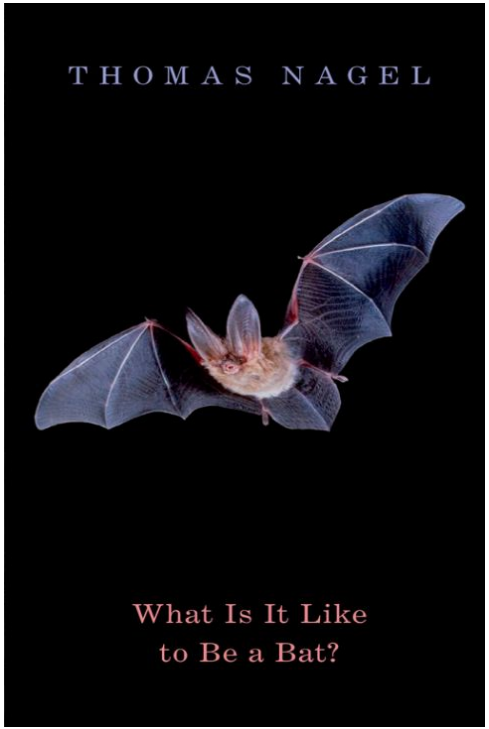
I am building to understand



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Phenomenology

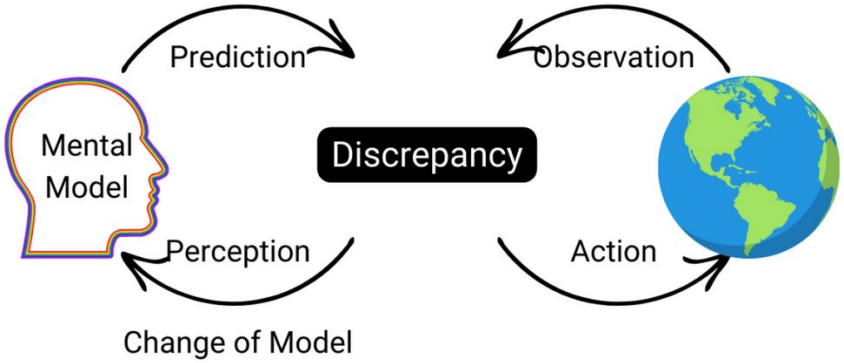
Phenomenology seeks to determine the *essential properties and structures of experience itself*, and to provide an *objective* account of subjective experience



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Active Inference

An enduring being actively infers the causes of its sensory input and shapes its environmental niche to minimize prediction errors



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Agency

Agency is the ability of a self to act purposefully to achieve its own goals in an uncertain world it experiences



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My project is to program a robot so it learns to survive by experiencing, engaging, and making sense of its world

All on its own

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Principles

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A self is mortal

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Agency implies a self distinct from an environment it depends on *to survive*

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A self is not a “thing” but processes organized to endure

A self is a dynamic organization of processes that constrain and sustain one another

The organization is not fixed but adapts to avoid dissipation

To adapt, a self must continually model itself in its world

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A self's overarching goal is to survive

The processes producing the self must adapt their organization to preserve *continuity of being*

A self strives to remain a self, not necessarily stay the same

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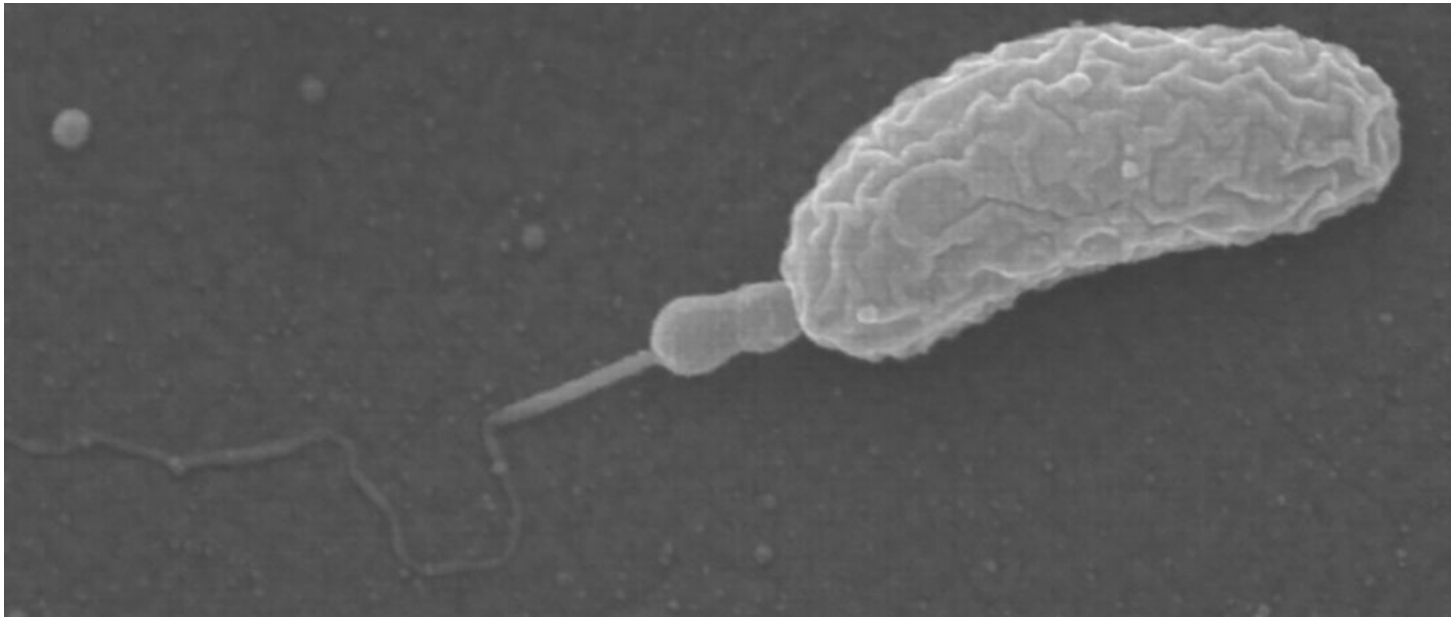
Mortality introduces value

What helps the self persist is good for it

What threatens to terminate it is bad for it

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Value is subjective



When a bacterium feeds on another, what's good for one is bad for the other

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This project defines **agent** as

“A mortal self who chooses how to survive”

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To survive, an agent must make (enough) sense of itself in its environment to predict (well enough) the consequences of its actions or inaction

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An agent has felt experiences

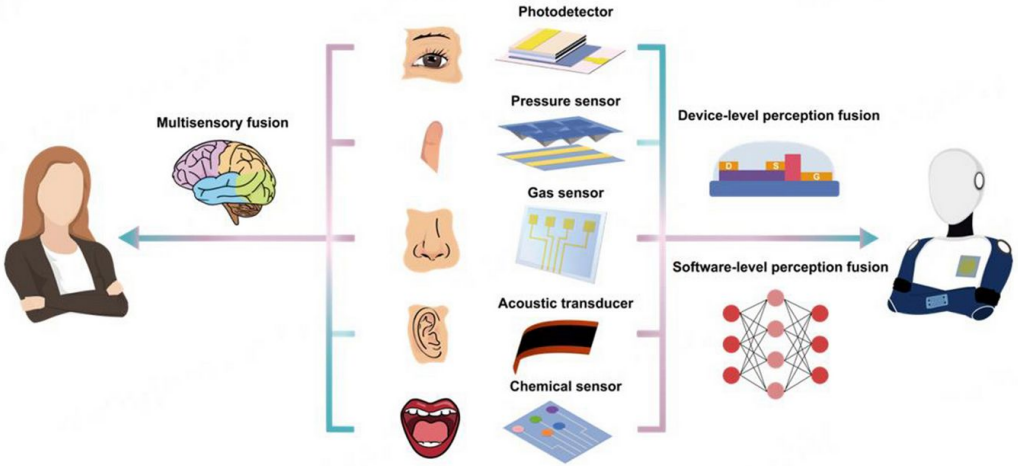
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The agent's environment impinges on its senses

So do the agent's own actions

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An agent constantly processes streams of multimodal sensations



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The agent integrates sensory streams in spacetime to create its own **experiences**

When I bite into an apple, I experience its taste, crunchiness, color, weight etc. at once, in unified moments of awareness

An agent need **not** be aware it is having experiences in order to have them

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The agent keeps track of risks to its survival by constantly measuring its **wellbeing**

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The wellbeing of an agent is a function of…



Fullness
(how energetic it is)



Integrity
(how healthy it is)



Engagement
(how participatory it is)

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The agent *correlates* its experiences with fluctuations in its wellbeing

This is how it has *felt* experiences

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An agent can have good and bad experiences



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An agent is sense making

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An agent must make sense of its world if it is to survive in it

An agent makes sense of its experiences by **modeling** what causes its sensations

It builds a **causal theory** to explain its sensations

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The agent uses its causal theory to make **predictions**

The causal theory is a *generative model*

It generates (predicts) incoming sensations including the sensory consequences of the agent's actions

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Actual sensations that contradict predicted sensations become **prediction errors**

Errors happen because the causal theory is inaccurate or incomplete, or because the world has changed since the causal theory was formulated

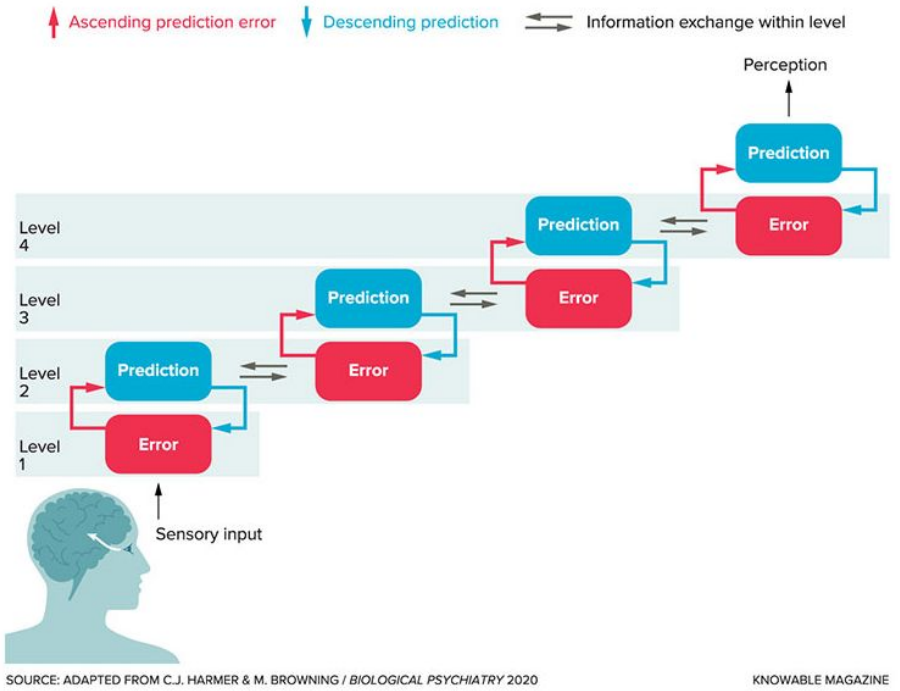
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Sensations = predictions + prediction errors

What an agent senses are the uncontested predictions it makes plus the prediction errors it receives

What the agent senses is limited to what it can predict

What it can predict changes over time



https://www.brainfacts.org/thinking-sensing-and-behaving/thinking-and-awareness/2021/psychedelics-open-a-new-window-on-the-mechanisms-of-perception-070821

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Surprising experiences

An **experience** is the unification of multi-modal sensations

An experience is surprising when it is composed of sensations that are prediction errors



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At first, an agent holds a trivial causal theory that simply predicts unchanging sensations

Expecting a world that never changes leads quickly to prediction errors

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The agent updates its causal theory when too many experiences are surprising

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The agent uses its causal theory to predict incoming sensations but also to predict what actions favor good experiences over bad ones

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Attention sets the goals

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Not all experiences are worthy of being acted on

The agent pays **attention** to the experiences that matter most to its survival, those it feels most intensely

Not only the ones that feel worse but also the ones that feel best

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The **confidence** an agent has in its experiences modulates how it pays attention to them

The agent assigns itself the **goals** of impacting experiences it pays attention to

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An agent seeks to gain confidence in its experiences before attempting to persist good ones and terminate bad ones

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An agent has confidence in experiences that are *not* surprising or, if they are, that are *enduring*

Even if an experience is surprising, the longer it persists the more confident the agent is about it

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Prioritizing existential goals

An agent prioritizes goals that can improve its wellbeing



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Pursuing exploration goals

When survival is not at stake, an agent can act for the sake of new experiences

It babbles



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Implementation

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Project challenges

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Somehow, the robot must be a **mortal agent**

So it can have *good vs bad* experiences and take *beneficial* actions

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Somehow, the robot must **experience** its world, **make sense** of its experiences, and **act** to promote good ones

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Somehow, the robot's experiences, sense making and goals must be ***entirely its own***

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What of Active Inference?

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Active Inference does not tell me *how to program* a mortal robot that actively learns how to survive

It tells me *what information dynamics are required* for the robot to do so

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Active Inference sets up **requirements**

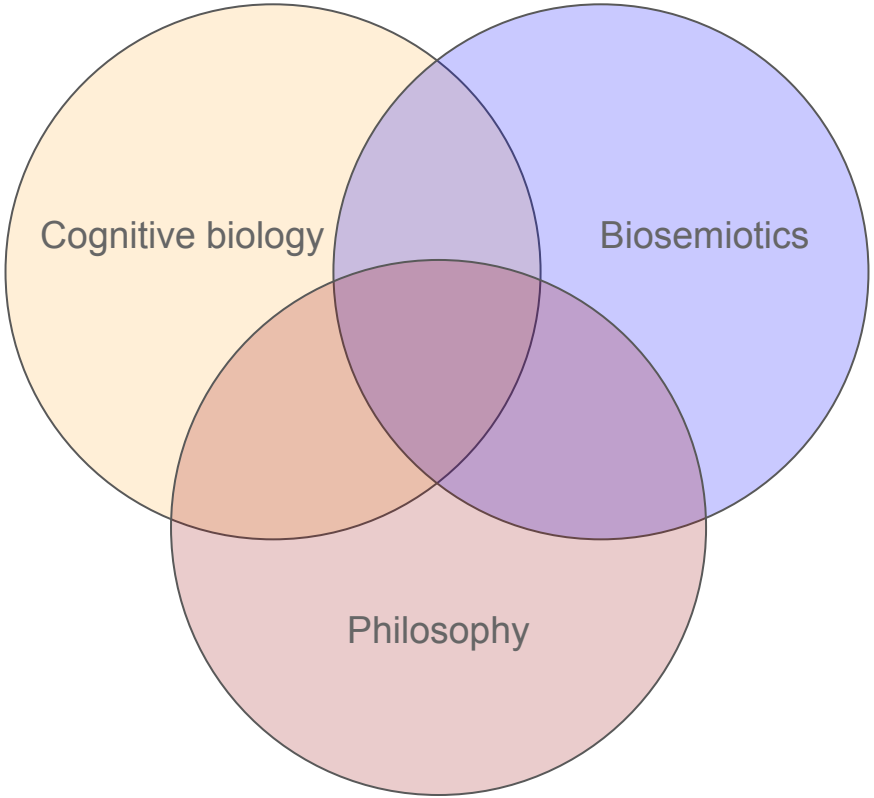
The observable states of the robot must conform to the *expectations* set forth by the Free Energy Principle

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Building a robot’s mind

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The implementation draws inspiration from…



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Ideas from cognitive biology

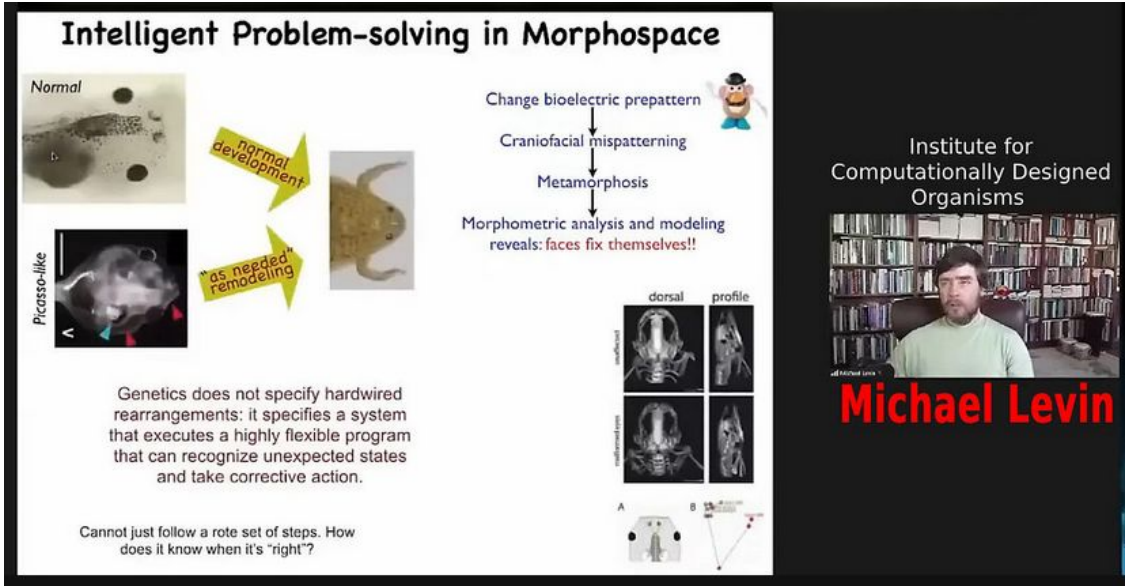
Organism as mutually sustaining/constraining processes

Multiscale competency

Stress as cognitive glue

Problem-solving in morphospace

Cognitive light cones



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Ideas from biosemiotics

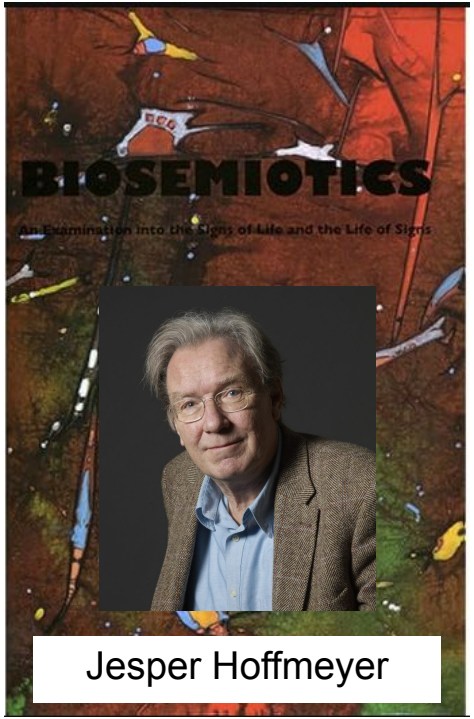
Umwelts

Aboutness

Signal processes

Sense making





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Ideas from philosophy

Kant’s Synthetic Unity of Apperception

Google DeepMind’s Apperception Engine



Richard Evans

Deep Mind

The Apperception Engine

Talk outline: Imagine a machine, equipped with various sensors, receiving a stream of sensory information. Somehow, it must make sense of that sensory stream. But what, exactly, does “making sense” involve, and how can it be implemented in a machine? I argue that making sense of a sensory stream involves constructing objects that persist over time, with properties that change over time according to intelligible universal laws (represented as logic programs). Thus, making sense is a form of program synthesis - but it is unsupervised program synthesis from raw unstructured input. I will describe our system, the Apperception Engine, that was designed to make sense of raw sensory input by constructing programs that explain that input. I will show various examples of our system in action. This talk will draw on two of our recent papers published in the Artificial Intelligence Journal: “Making Sense of Sensory Input”, and the sequel “Making Sense of Raw Input”.

Bio: Richard Evans is a Staff Research Scientist at DeepMind and an Honorary Senior Research Fellow at Imperial College. He is particularly interested in program synthesis, neuro-symbolic systems, and the philosophy of Immanuel Kant.

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The **software** is built using Elixir, Prolog and CHR

Elixir is a process-oriented and functional programming language

Prolog is a logic programming language

CHR (Constraint Handling Rules) is a constraint solving extension to Prolog

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The **hardware** is the robot's devices plus a high-end server

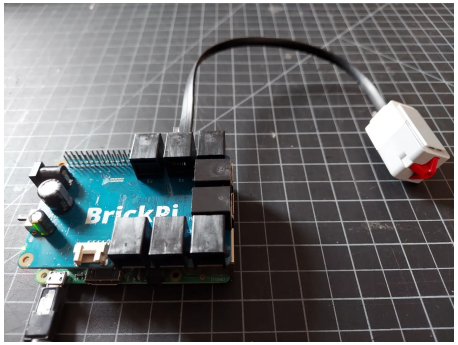
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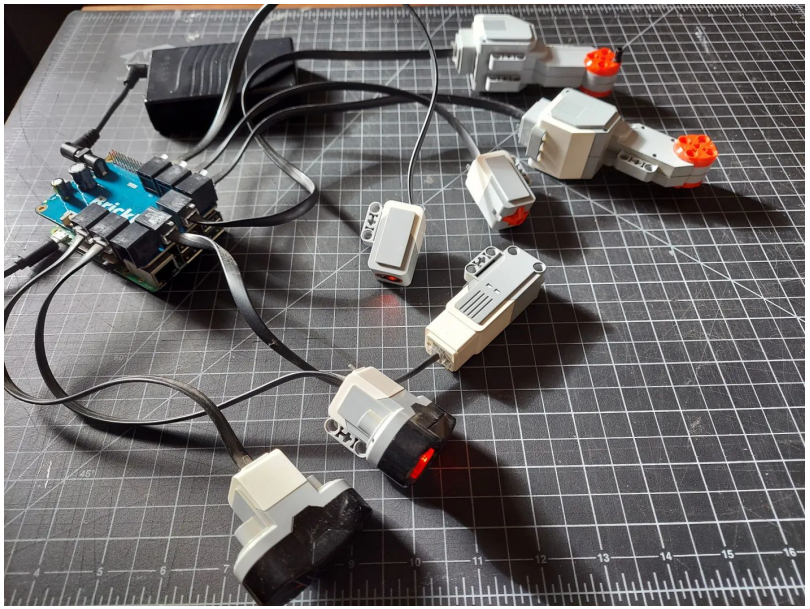
Robot body hardware

Lego EV3 sensors and motors

A Raspberry Pi3

A Dexter Industries BrickPi board to interface with the EV3 sensors and motors





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Robot mind hardware

On the backend, a 128-core server does the compute-intensive cognitive processing



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Writing mortal code

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Can a robot be made mortal?

The robot's physical structure is immortal but the *processes* animating it come in and out of existence

A robot’s *mind* can be made mortal

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The robot grows a *mortal mind* to animate an *immortal body* so it can survivably explore/exploit its world

The robot starts with a minimal mind and a reserve of wellbeing

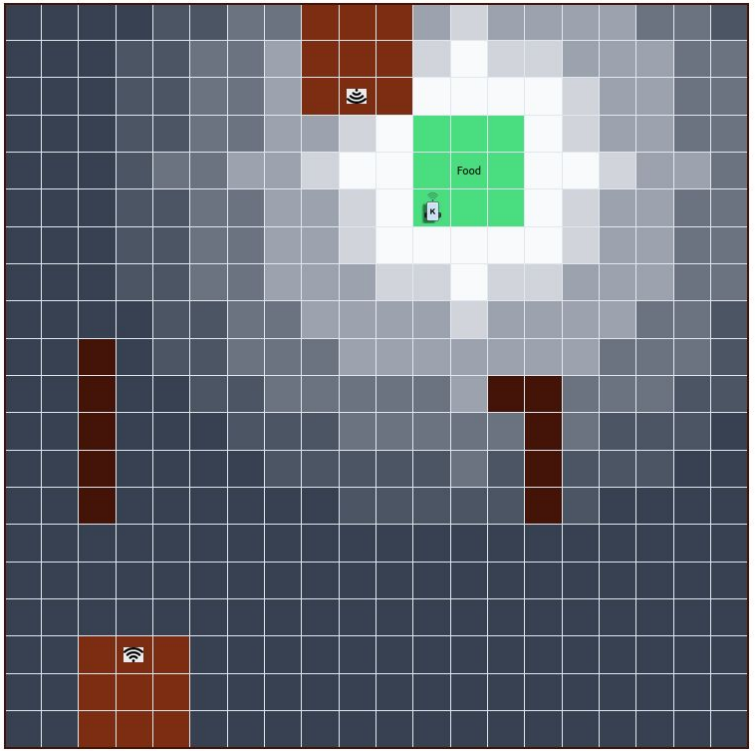
It must learn to maintain sufficient wellbeing or else its mind dies

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Virtual grid world

A virtual grid world provides a virtual robot with an environment that can both nourish (food) and damage it (obstacles, poison)

A physical environment will be built that recreates the conditions of the virtual world but with the messiness of the analog world

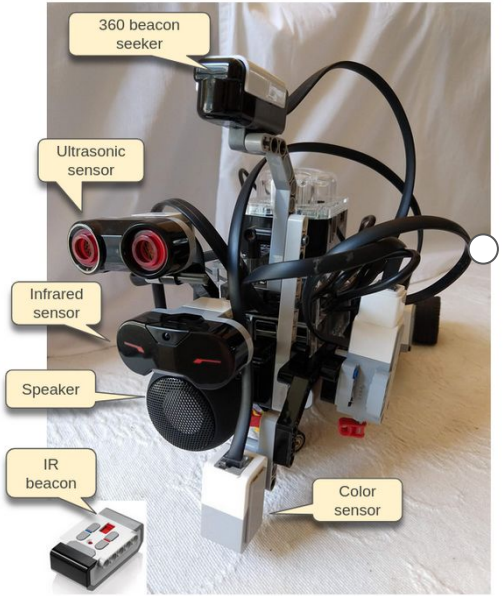


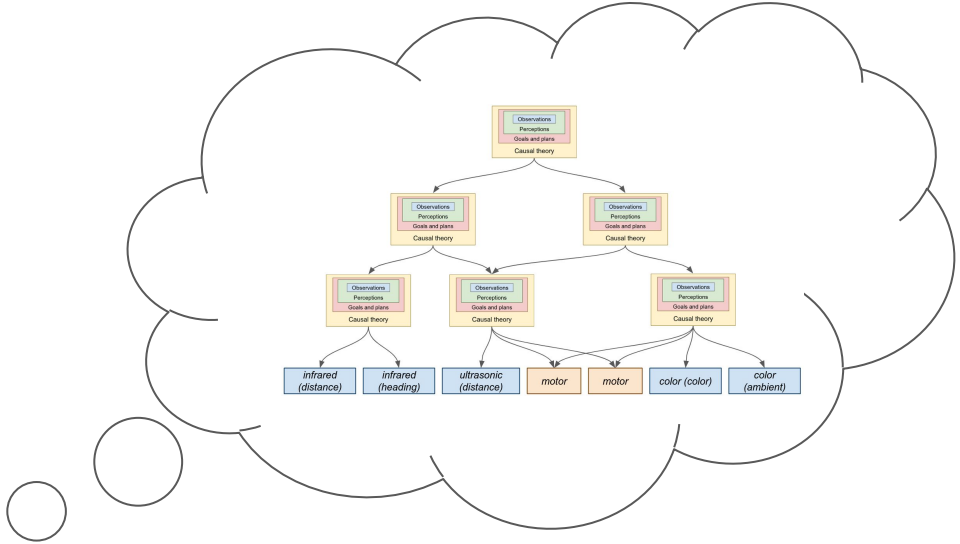
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Growing the robot’s mind

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The robot's mind is an evolving collective of long-lived processes





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All intelligence is collective intelligence

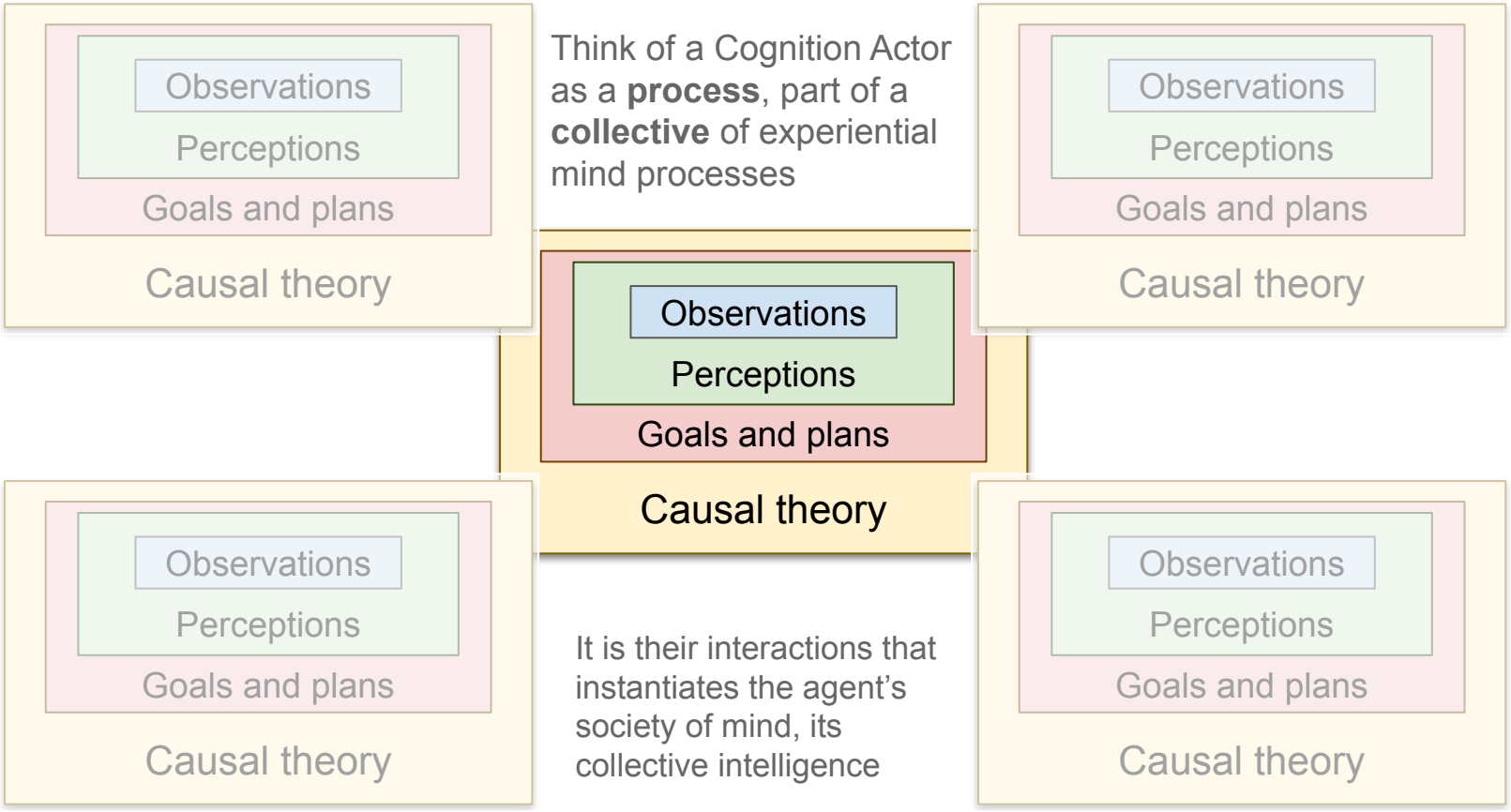
The robot is not a monolithic agent

It contains multitudes

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The robot's mind is implemented as a dynamic **collective** of **cognition actors (CA's)**, each realizing a piece of the robot's overall experience

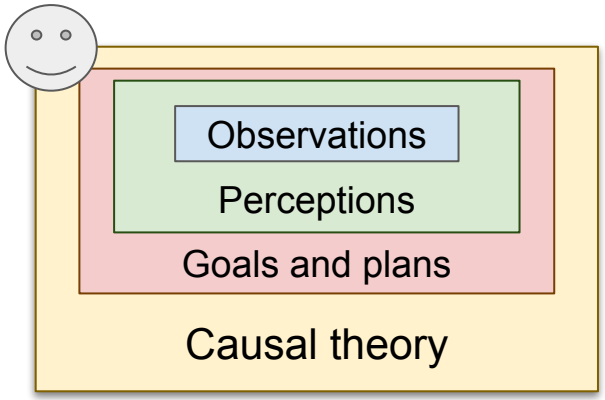
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Each cognition actor (CA) is implemented as a *long-lived, stateful process* that repeatedly

- makes **observations** from predictions and prediction errors
- builds a **causal theory** to make sense of these observations
- **unifies** patterns of observations into **perceptions**
- assigns **value** to them
- **acts** to cause desirable perceptions



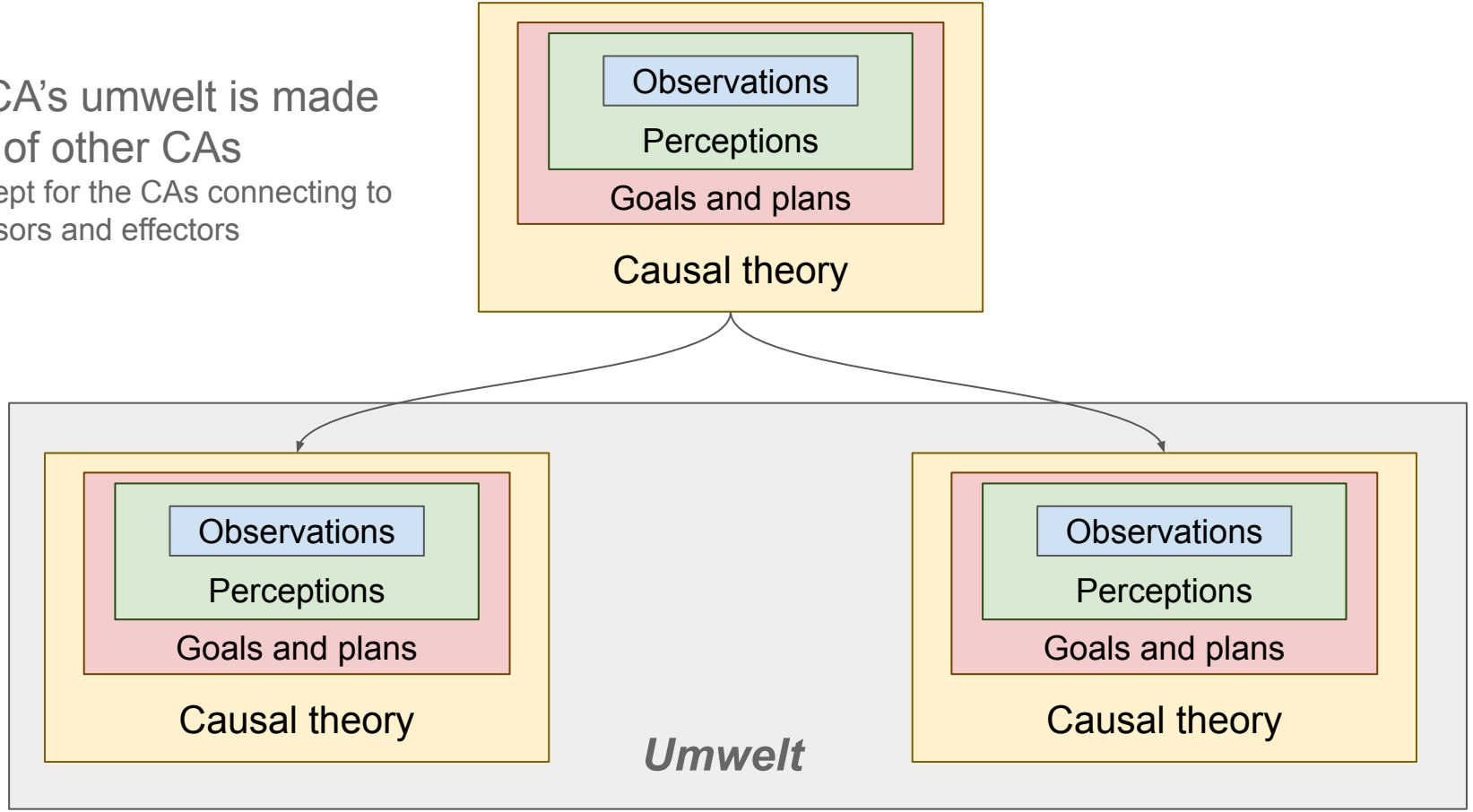
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Each CA has its own **umwelt** (its self-centered world) that it observes, makes sense of and acts on

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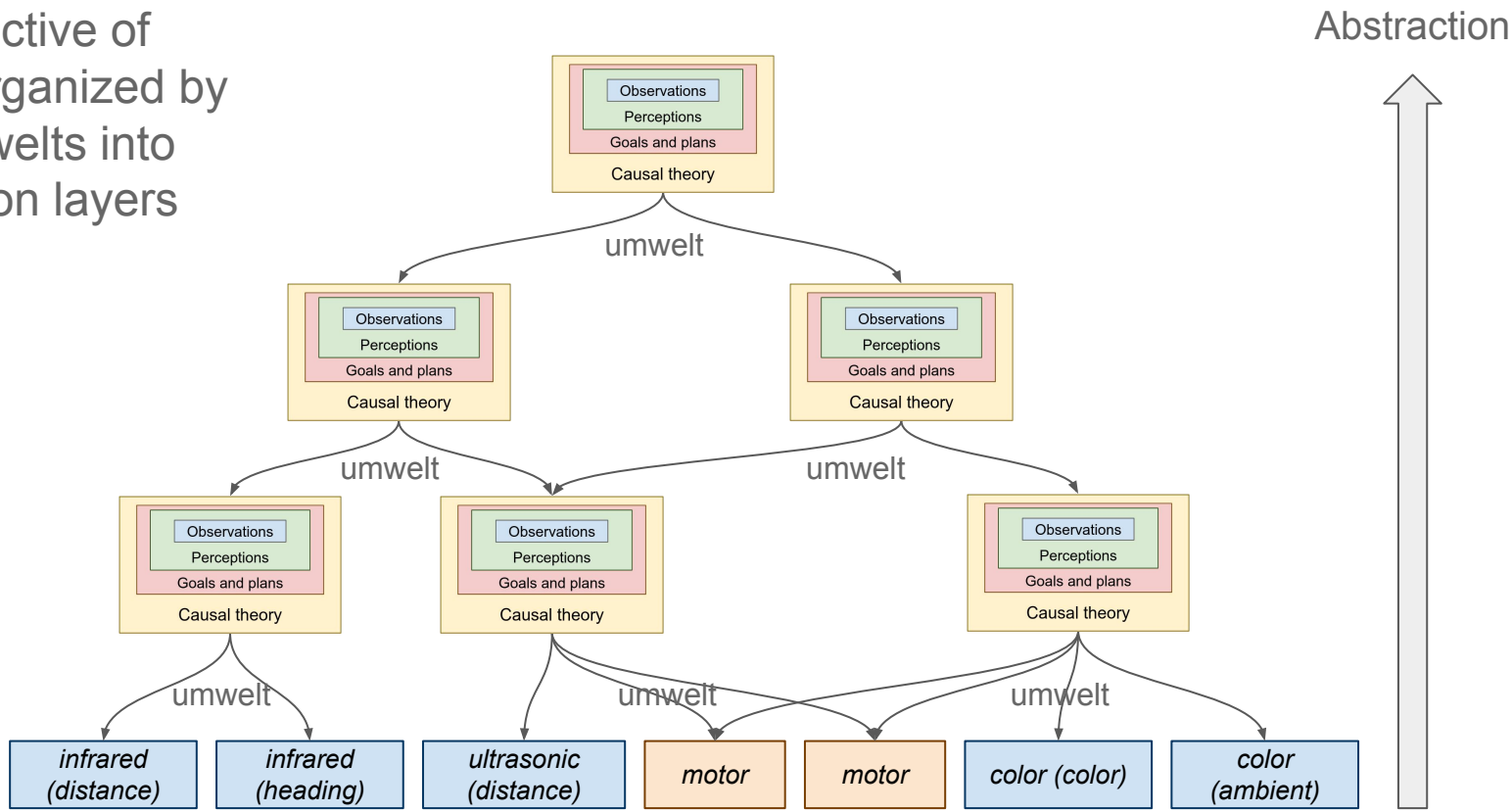
A CA's umwelt is made up of other CAs

except for the CAs connecting to sensors and effectors



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The collective of CAs is organized by their umwelts into abstraction layers



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The bottom layer is populated a priori by **static CAs**

They are static only in the sense that they are permanent

Static CAs interface directly with the robot's sensors (sensor CAs) and motors (effector CAs)

Their umwelts is the robot's sensed, acted upon world



Static CAs



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All other layers of CAs come in and out of existence during the lifetime of the robot's mind

They capture what the robot learns from striving to survive

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Survival of a collective of processes

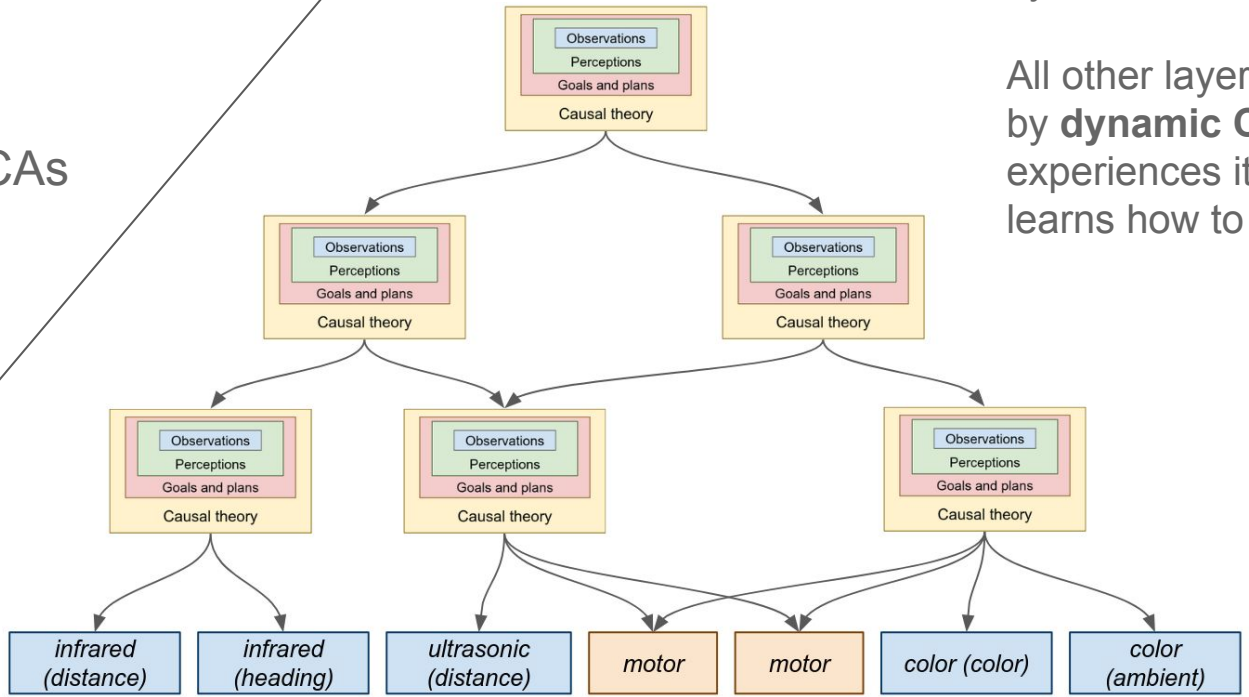
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The *a priori* bottom layer of static CAs is not enough by itself for the robot to find nourishment and avoid hazards

Additional layers of CAs are needed to create the “big picture” understanding and affordances needed for competency and survival

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Dynamic CAs



Static CAs

The first row is pre-populated by static CAs

All other layers are populated by **dynamic CAs** as the robot experiences its world and learns how to survive

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A society of mind grows, shrinks and reshapes itself by adding and removing dynamic CAs

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A society of mind shapes itself to capture what it learns as it strives to survive

It searches for a competent self-organization in the space of all possible “shapes” it could take (its morphospace)

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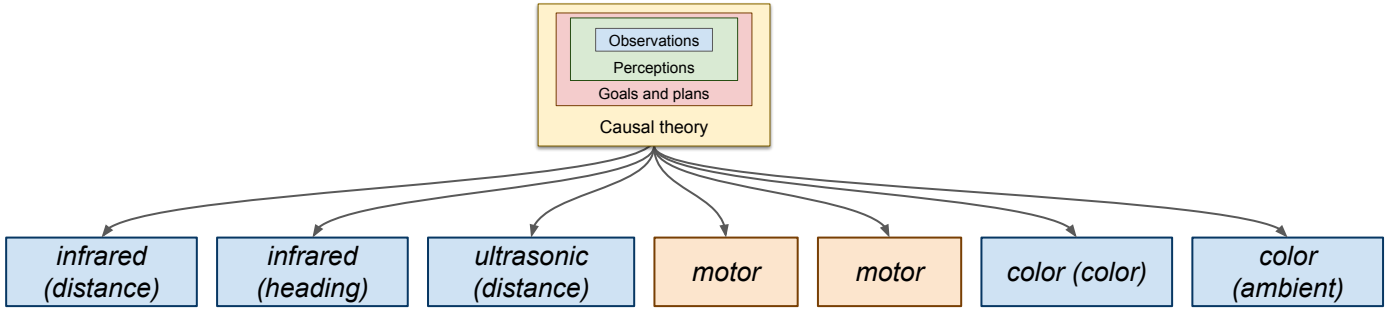
To help the search converge to a competent organization, the collective applies principles from developmental biology

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- Parts from wholes, not wholes from parts: Grow by replication, division and differentiation, not by assembly
- Use it or lose it: Grow to over-abundance then trim to essential
- Stress as cognitive glue: The stress felt by parts at risk propagates and motivates global change
- Stable wholes, transient parts: The simpler the parts, the faster they are replaced

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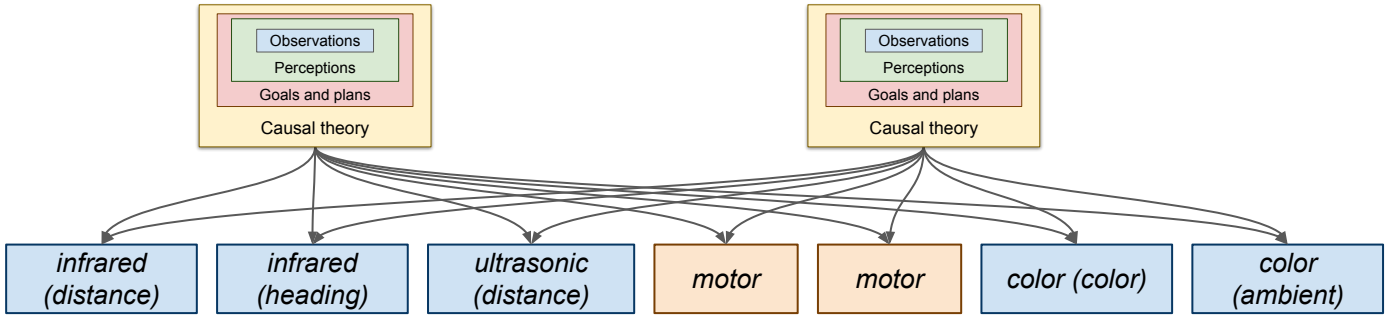
The first dynamic CA to occupy a layer grabs the whole layer below as its umwelt



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The dynamic CA then *replicates* until it saturates its own layer with replicants

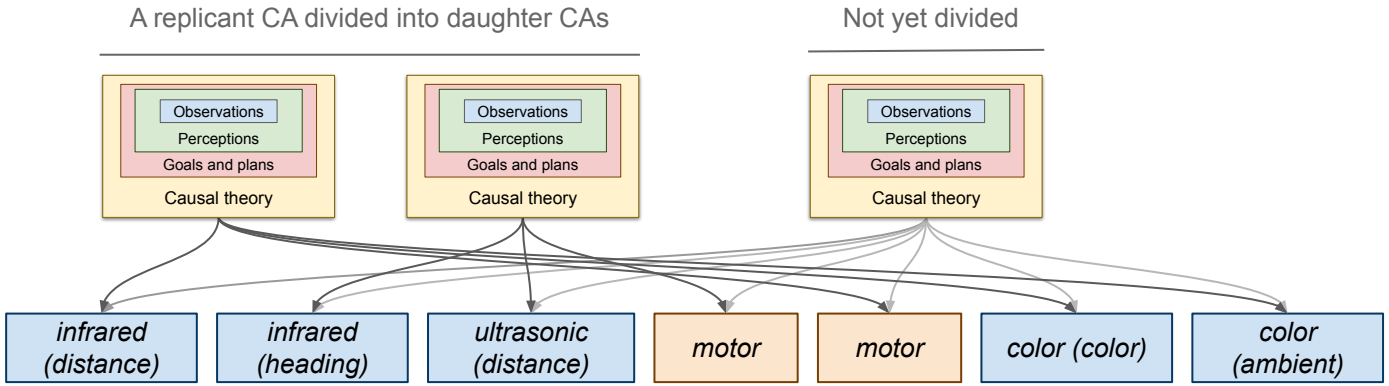
Replicants *differentiate* their internal states over time because life cycles are not deterministic



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After saturating their layer, replicant dynamic CAs begin to divide into daughter CAs

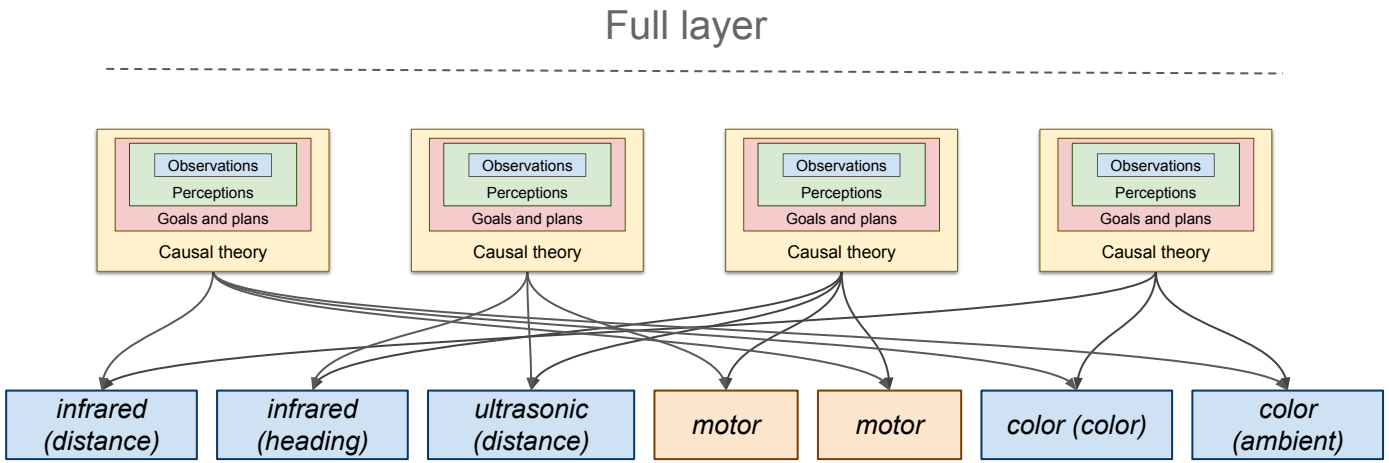
Each daughter CA gets a random half of its mother's umwelt



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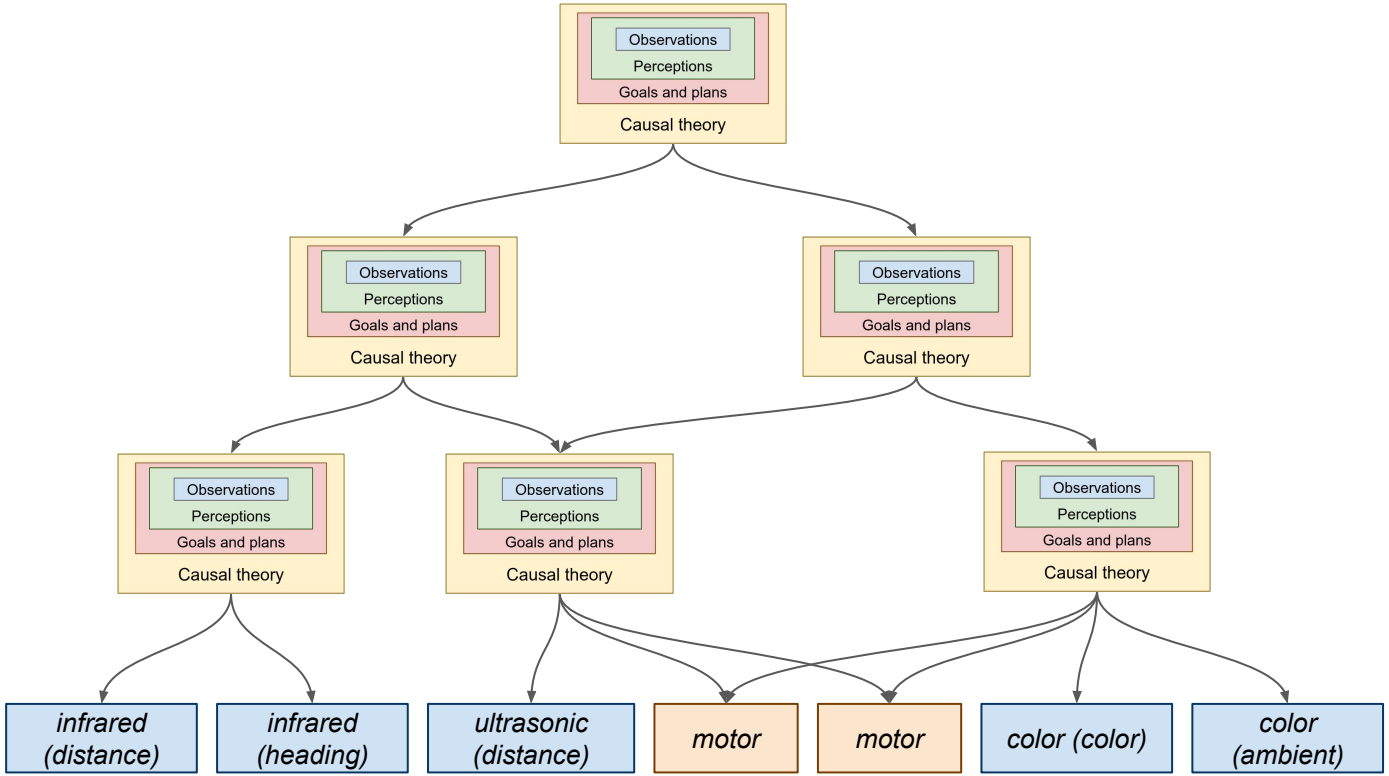
Dynamic CAs keep dividing until then can no longer do so

At that point, the layer is full and a new layer can be started



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Over time dynamic CAs grow into an abundant abstraction hierarchy



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A dynamic CA will remove itself if it runs out of fullness or integrity or if it persistently serves no purpose

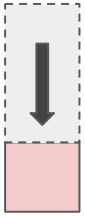
This removes useless CAs and can make room for other CAs to come into existence and perhaps do better

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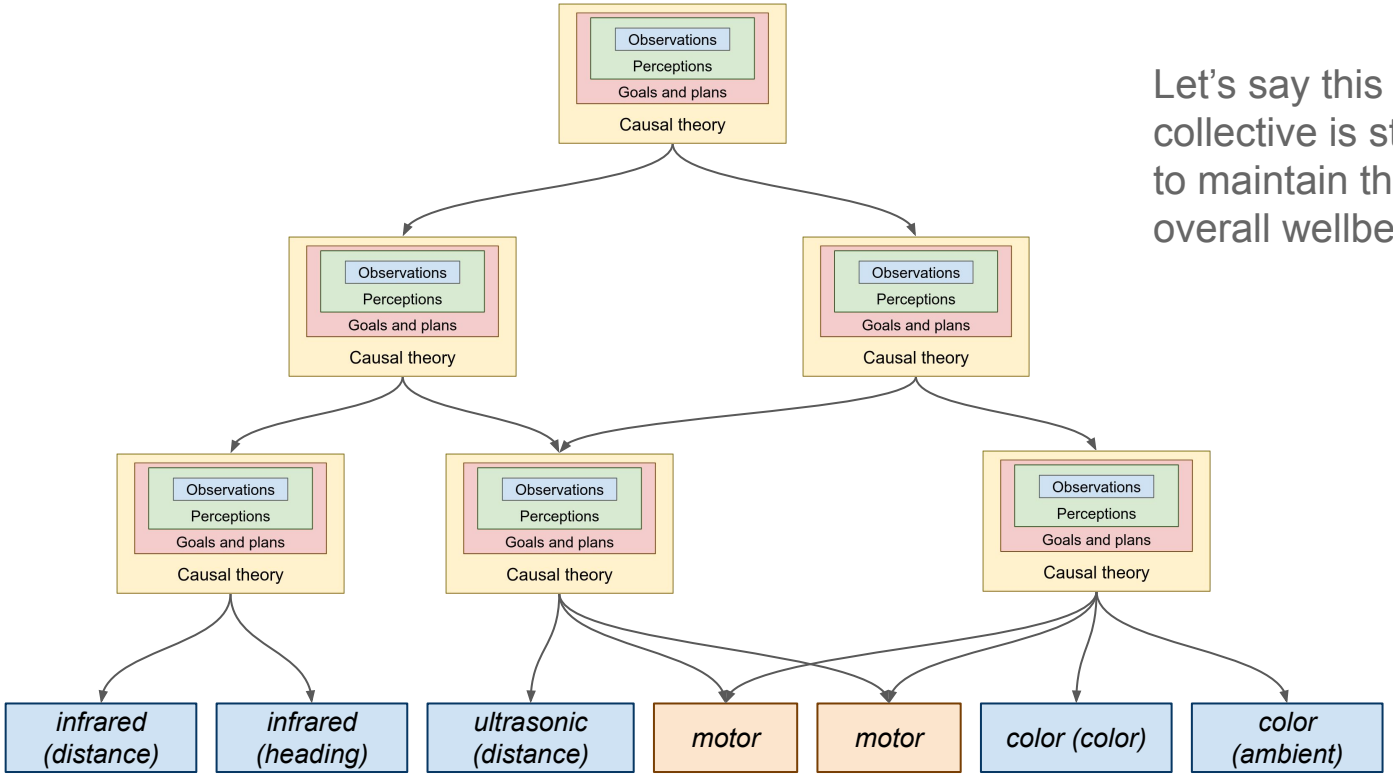
This adding and removing of CAs and layers of CAs explores a space of possible organizations of the mind

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Searching for a **competent** collective



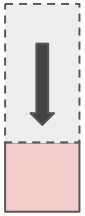
Wellbeing



Let's say this particular collective is struggling to maintain the robot's overall wellbeing

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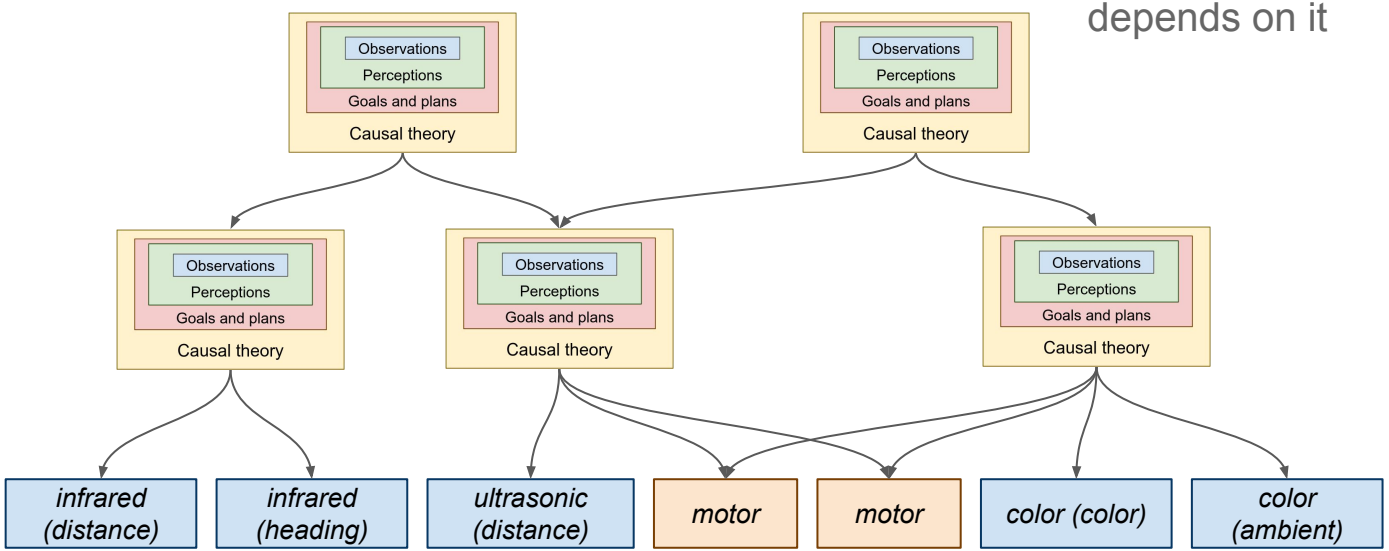
Searching for a **competent** collective



Wellbeing

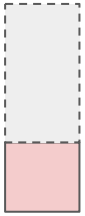
The top CA removes itself

It can because no other CA depends on it



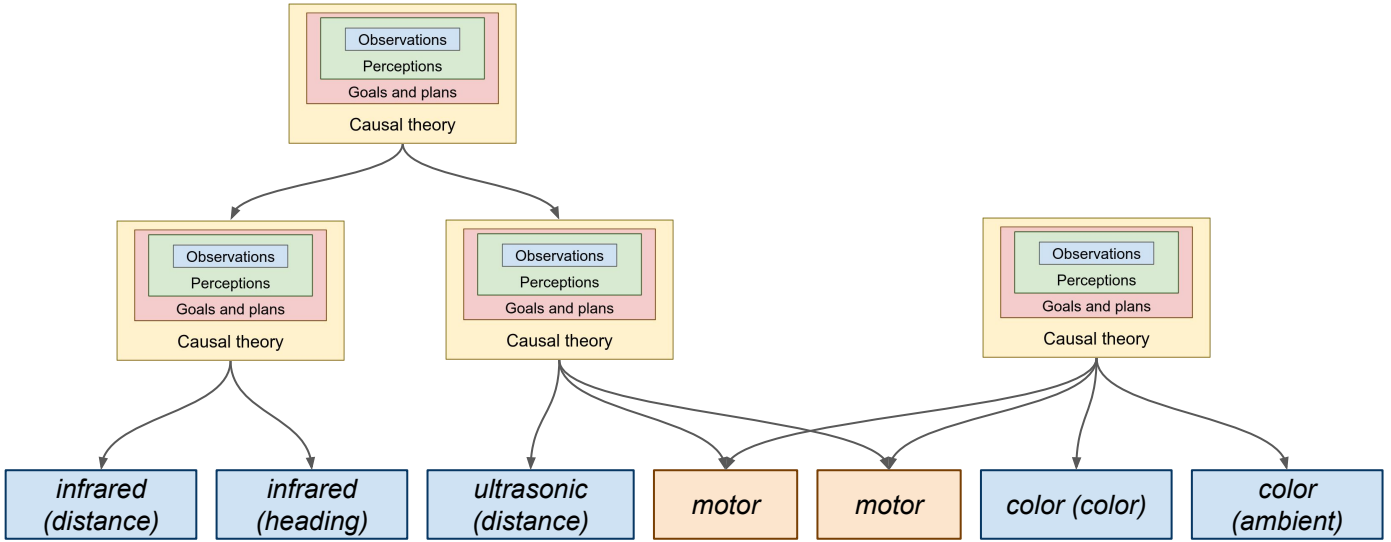
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Searching for a **competent** collective



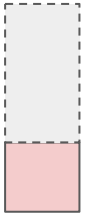
Wellbeing

This allows for other underperforming CAs to also remove themselves



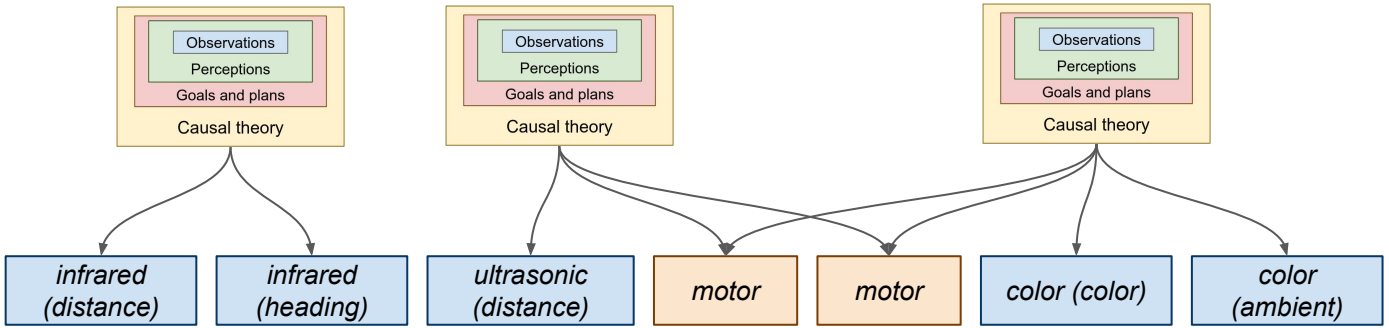
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Searching for a **competent** collective



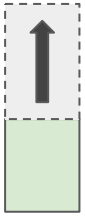
Wellbeing

And so on, reducing the “energy cost” and thus the loss of fullness by the collective



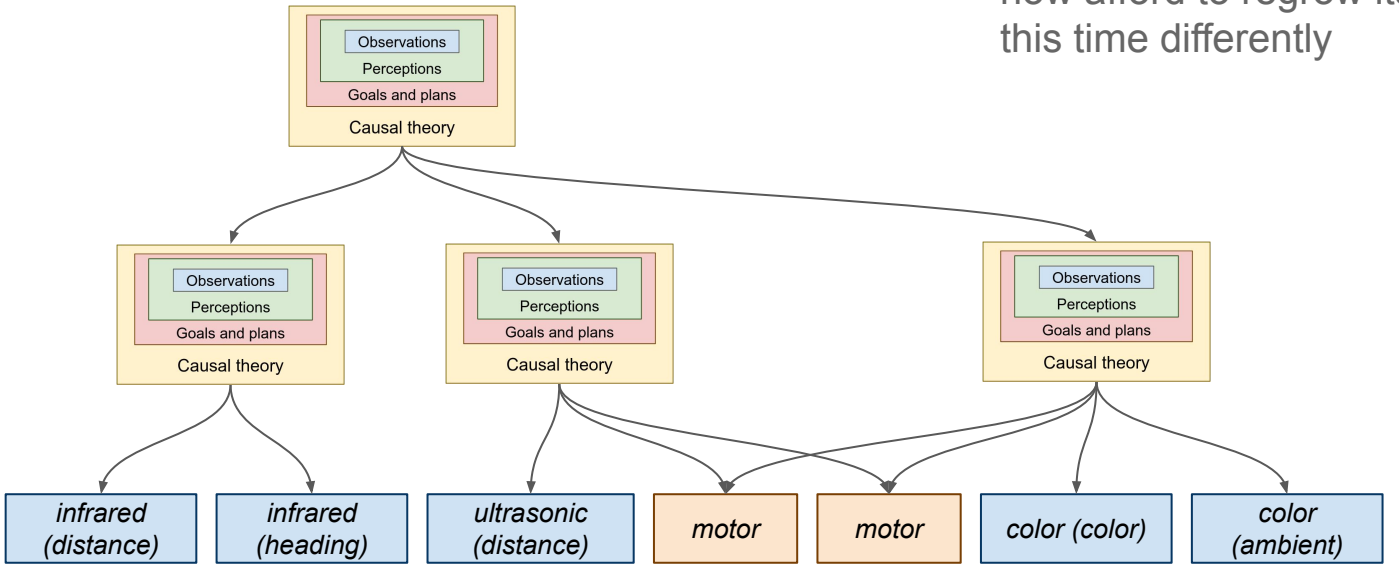
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Searching for a **competent** collective



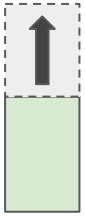
Wellbeing

Having a lower “energy draw”, the collective can now afford to regrow itself, this time differently

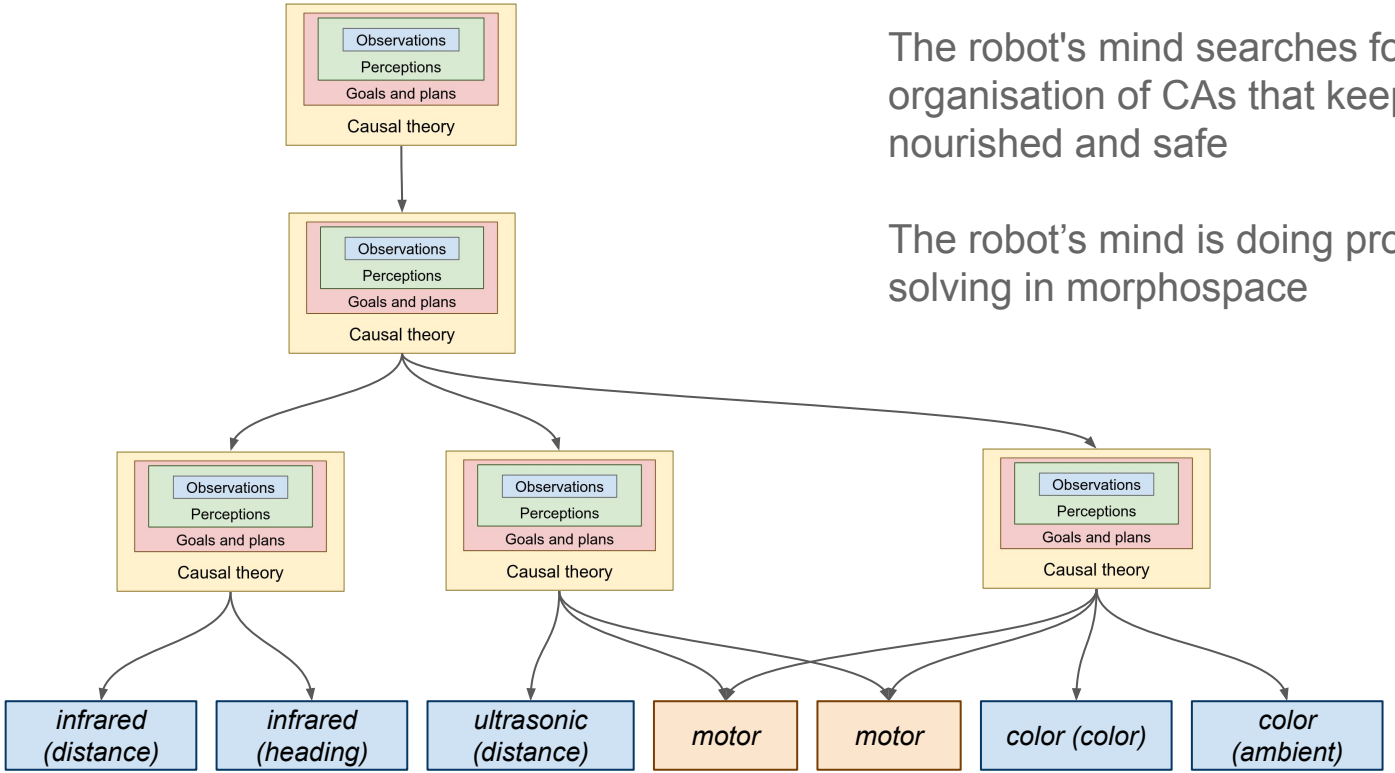


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Searching for a **competent** collective



Wellbeing



The robot's mind searches for an organisation of CAs that keeps it nourished and safe

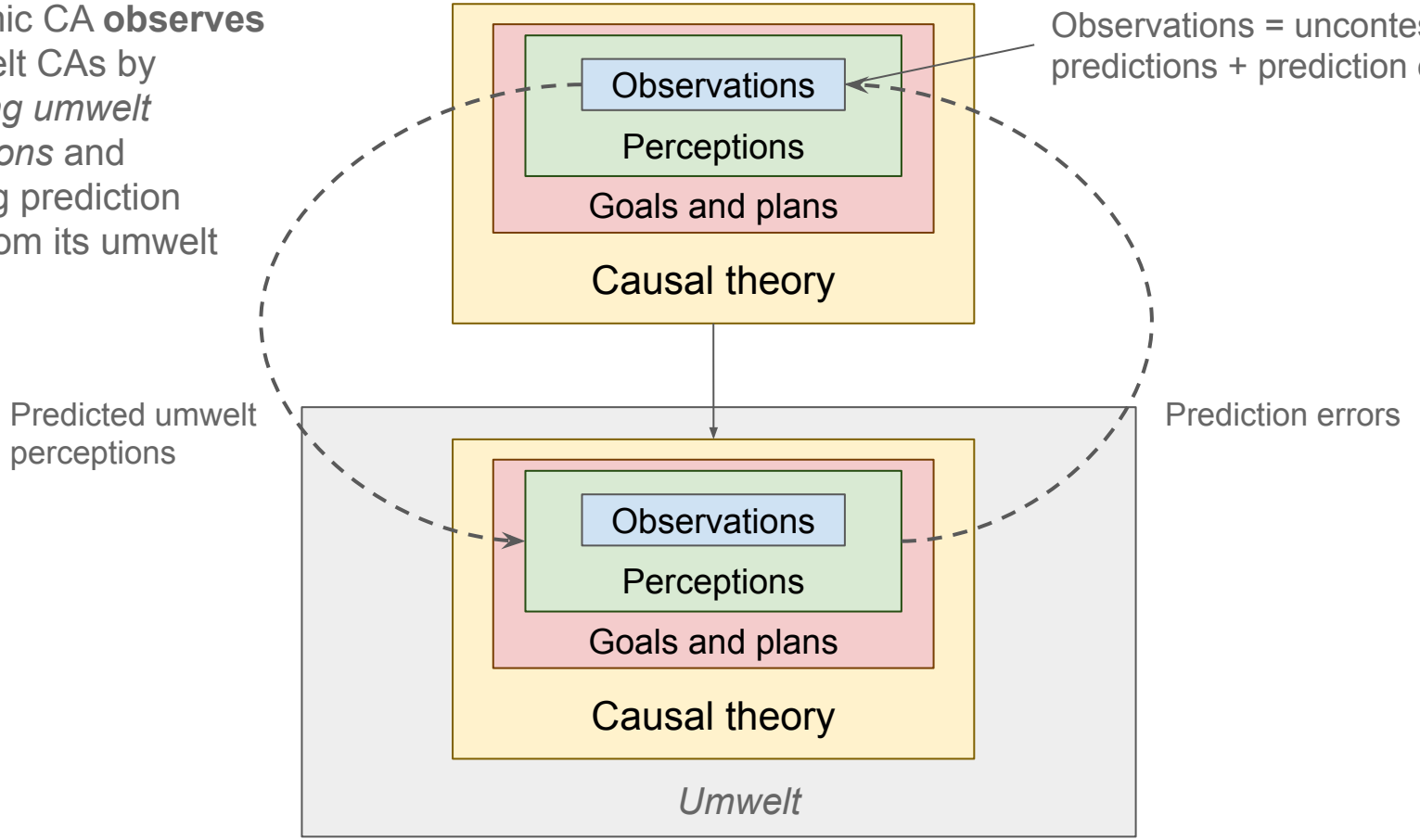
The robot's mind is doing problem solving in morphospace

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Observing and perceiving

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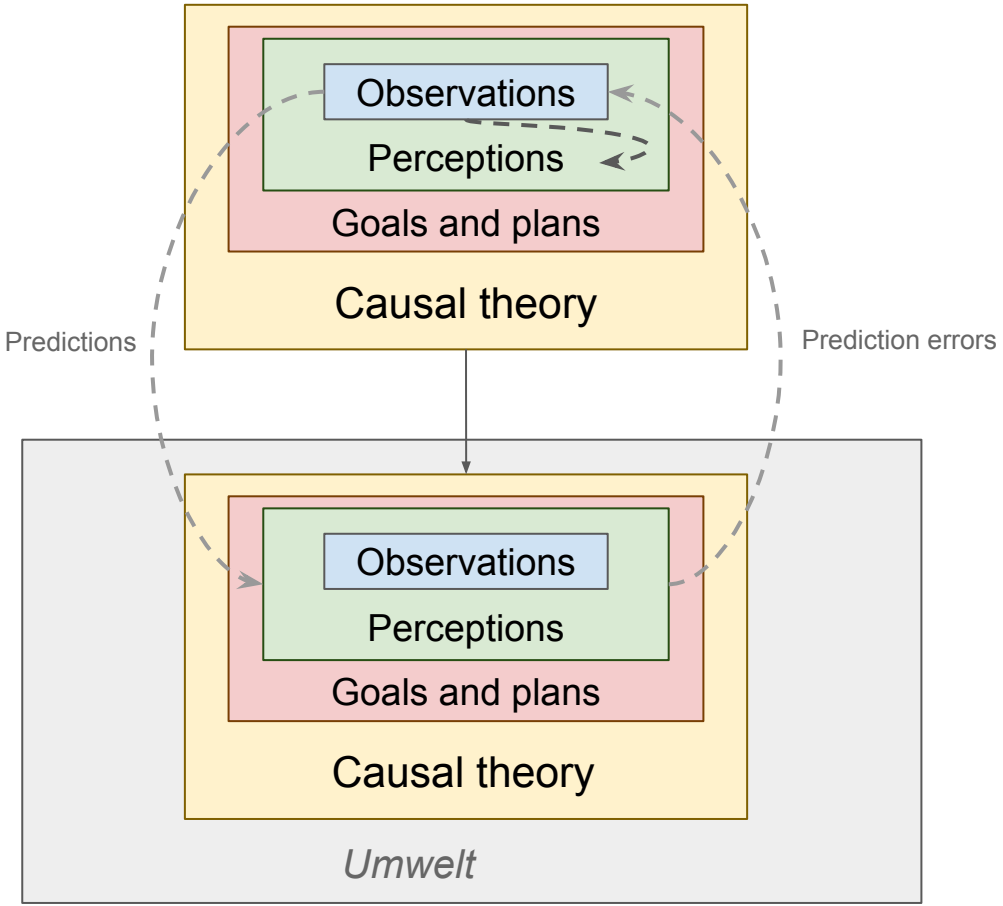
A dynamic CA **observes** its umwelt CAs by *predicting umwelt perceptions* and receiving prediction errors from its umwelt



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A dynamic CA *unifies* its past and present observations into **perceptions**

For example, having observed in its umwelt that distances keep shrinking, a dynamic CA would perceive "approaching"



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Sensor CAs (static CAs in the bottom layer) have simple perceptions from observing distance, color, luminance etc.

The observations of sensor readings are their perceptions

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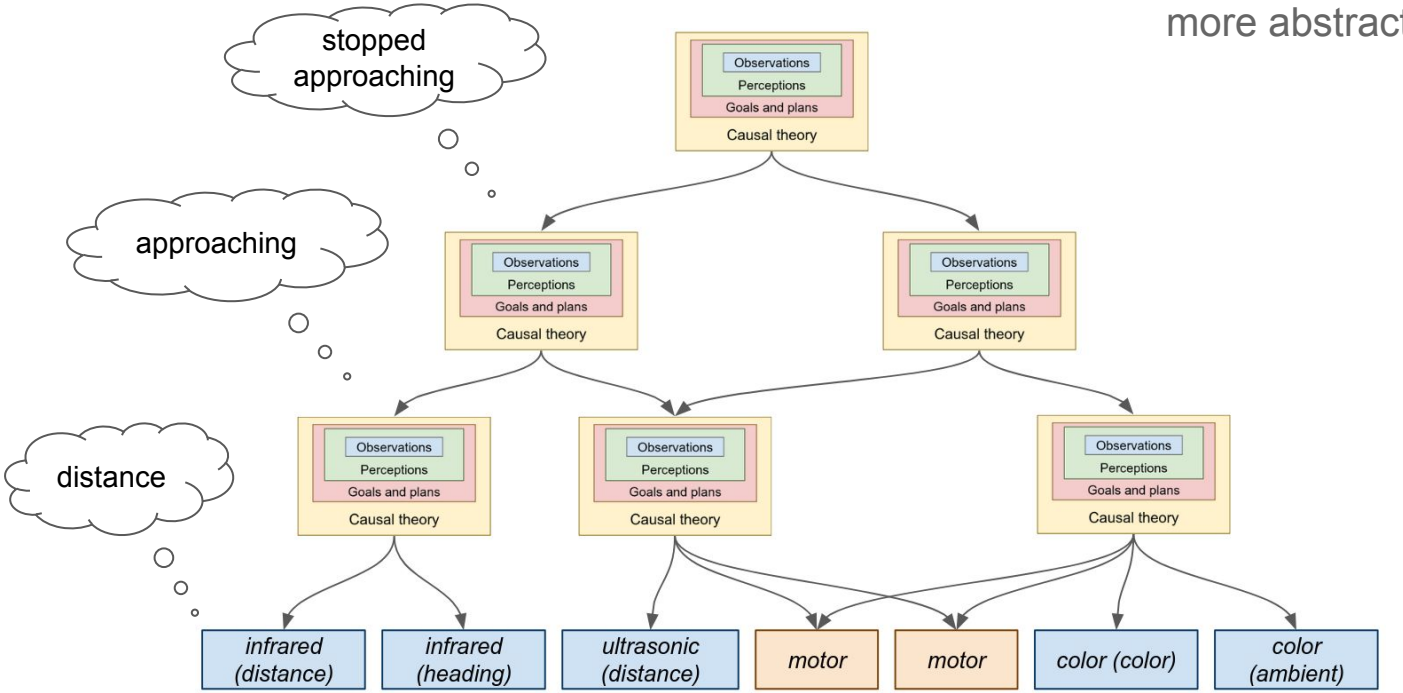
While a static CA observes and perceives the readings of its associated sensor, a dynamic CA perceives *patterns of perceptions* observed in its umwelt CAs

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Higher up dynamic CAs have more abstract perceptions than lower down CAs

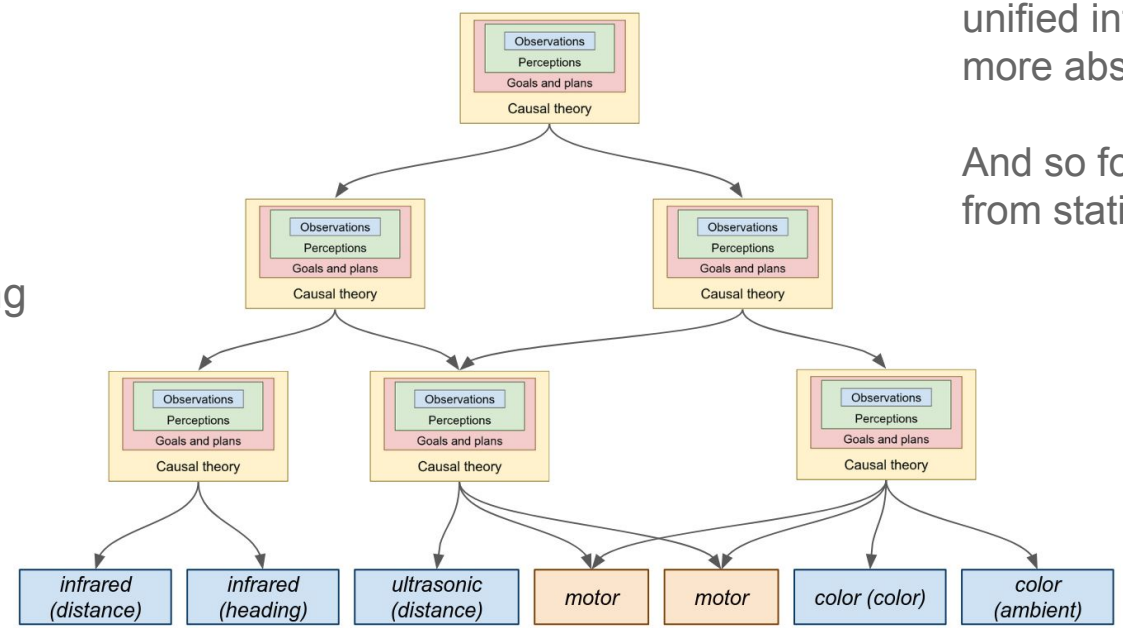
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For example, “stopped approaching” is more abstract than “approaching” which is more abstract than “distance”



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Perceiving



The perceptions of a CA become the raw material to be unified into the perceptions of more abstract CAs

And so forth, all the way up from static sensor CAs

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Perceptions are constructed and named by dynamic CAs, not by the programmer

They need not make sense to an external observer

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Feelings

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The CAs composing the robot’s mind feel the fluctuations in their individual wellbeing

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Everything a CA does increases or decreases its feeling of wellbeing

Any internal or external work reduces fullness

Spending time over a food patch increases fullness

Hitting an obstacle reduces integrity

Participating in an umwelt increase engagement, etc.

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Wellbeing measures are not separate

Activating motors is work that causes loss of fullness and integrity (wear and tear), but also increases engagement

Lost integrity is regained over time (think healing) by draining fullness, etc.

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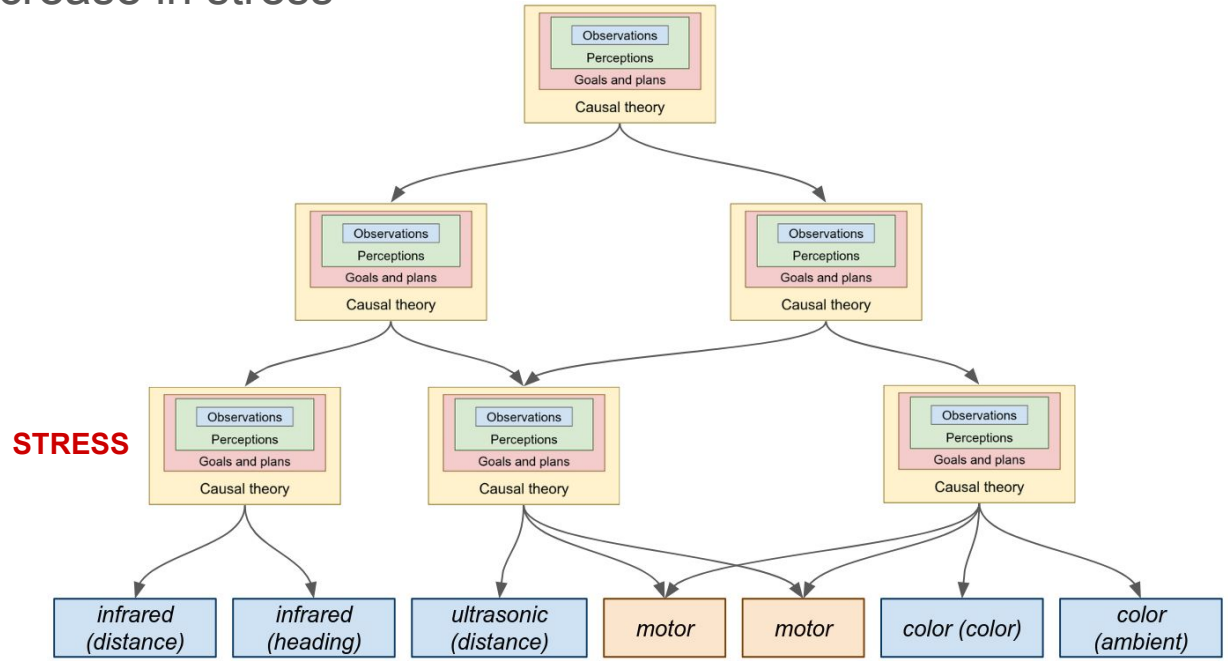
A CA's fluctuating wellbeing determines whether its *concurrent* perceptions feel good or bad

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The loss or gain in wellbeing of a CA diffuses throughout the collective as “stress” (increasing or decreasing)

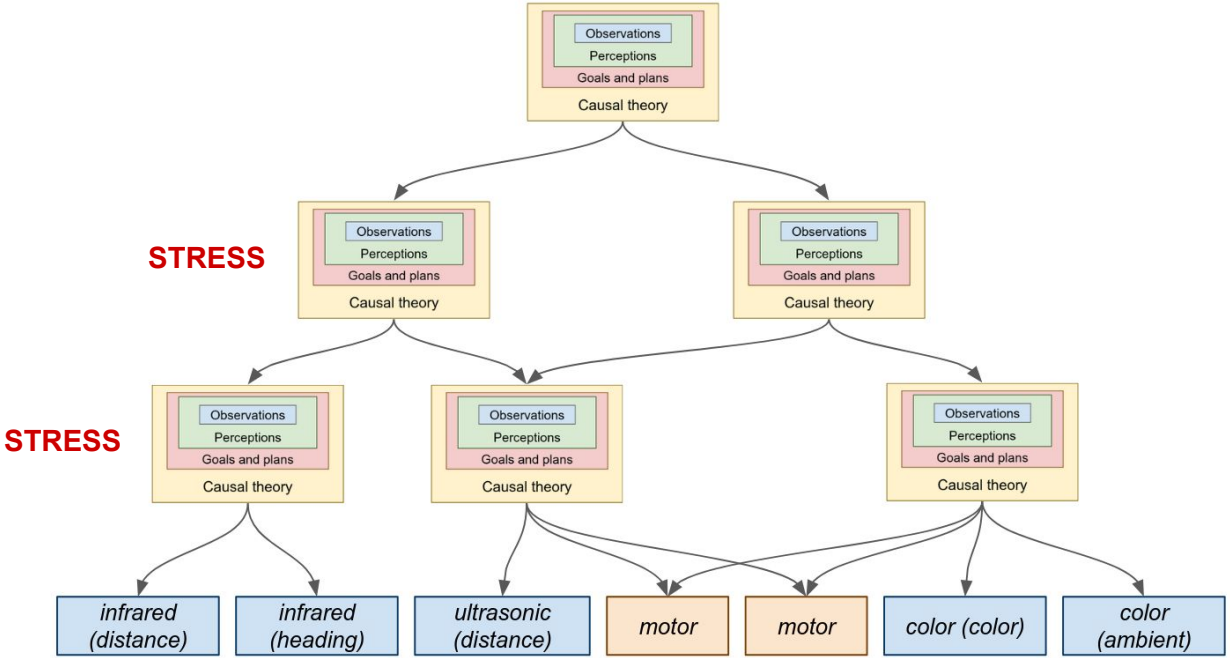
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Losing wellbeing causes an increase in stress



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Stress is contagious

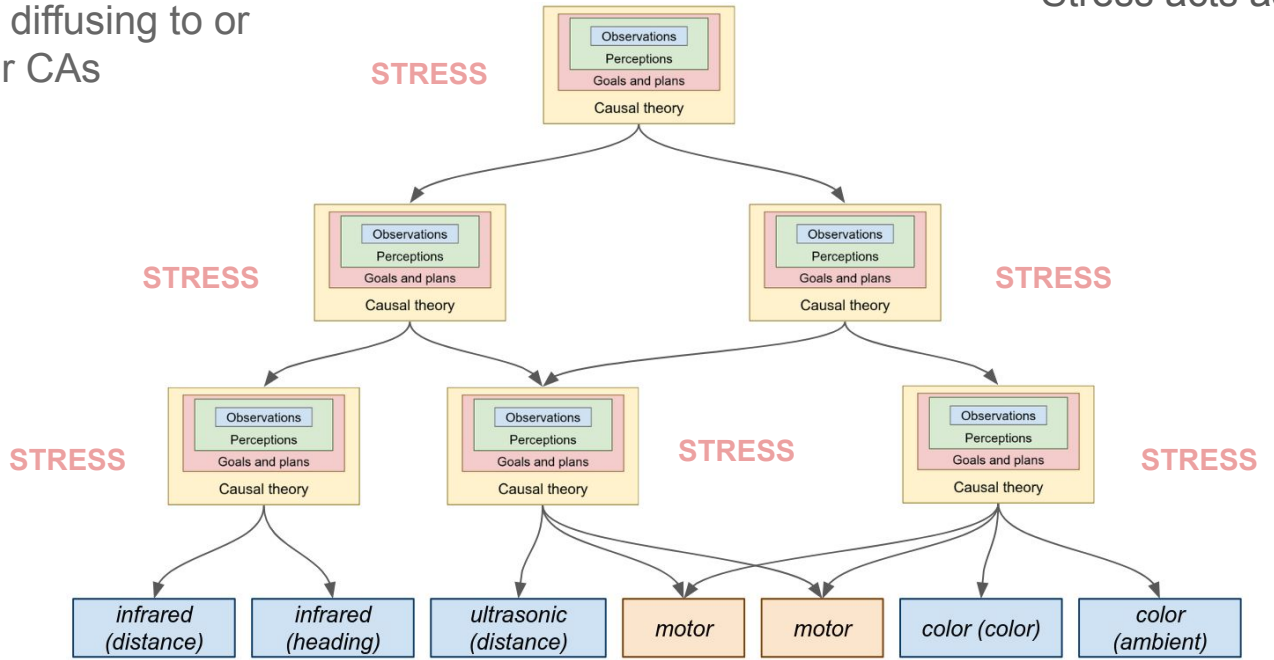


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A CA can not distinguish between a self-induced change to its wellbeing and a change caused by wellbeing diffusing to or from other CAs

This is how a CA's problem becomes every CA's problem

Stress acts as cognitive glue



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Stress alters the behavior of CAs

A “stressed” CA is more likely to try something different from what has worked in the past

A “comfortable” CA is more likely to settle into *habits*

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Acting

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A dynamic CA acts to confirm its perceptions, terminate its worse ones and persist the better ones

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A dynamic CA seeks to confirm, persist or terminate a perception it pays attention to

To do so, it must first identify the *observed* umwelt perceptions that it *unified* into the perception it now seeks to impact

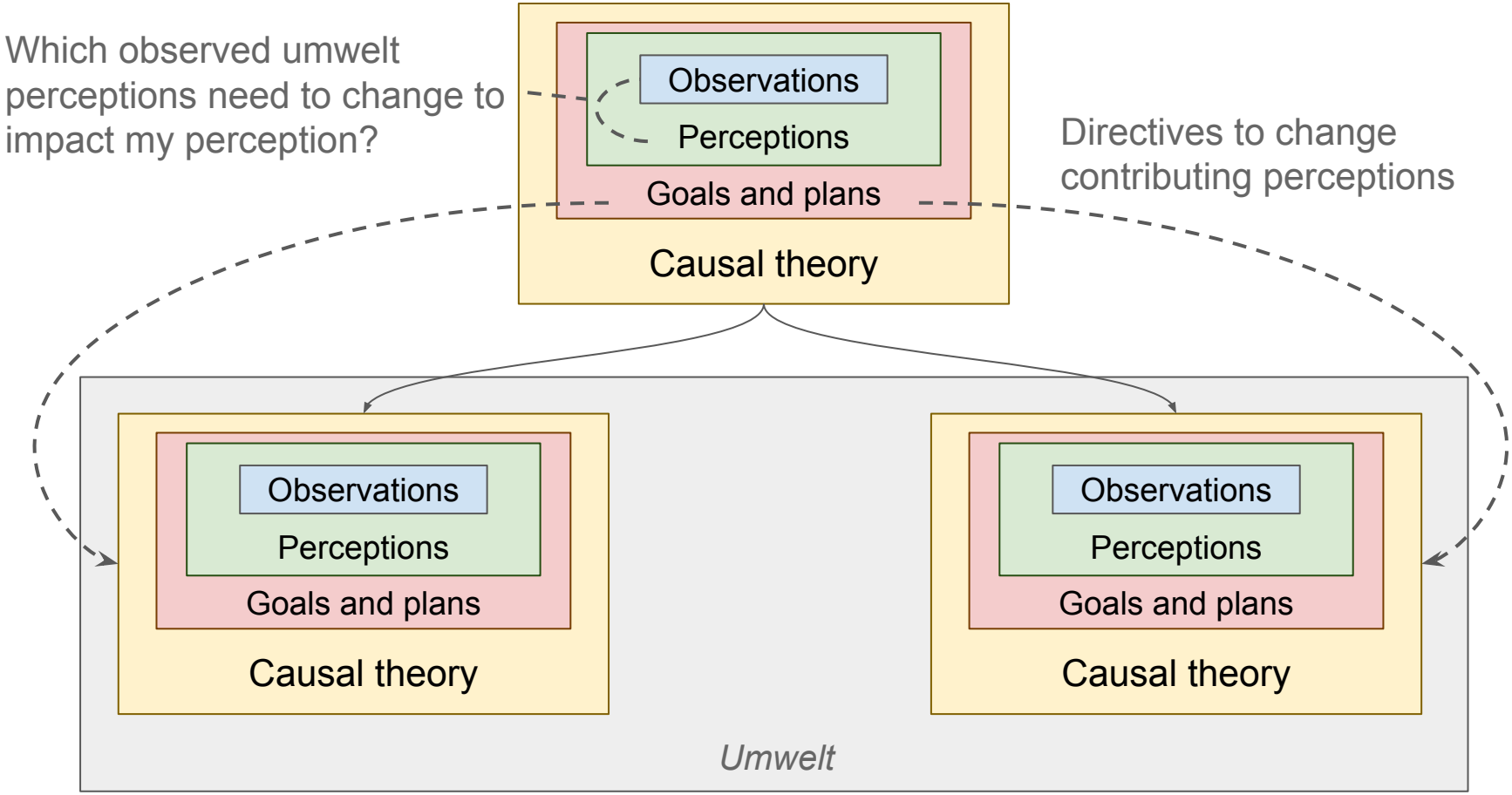
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The dynamic CA builds a **plan** to impact observed umwelt perceptions that were unified into its own perception

A plan is a set of **directives** it sends to its umwelt

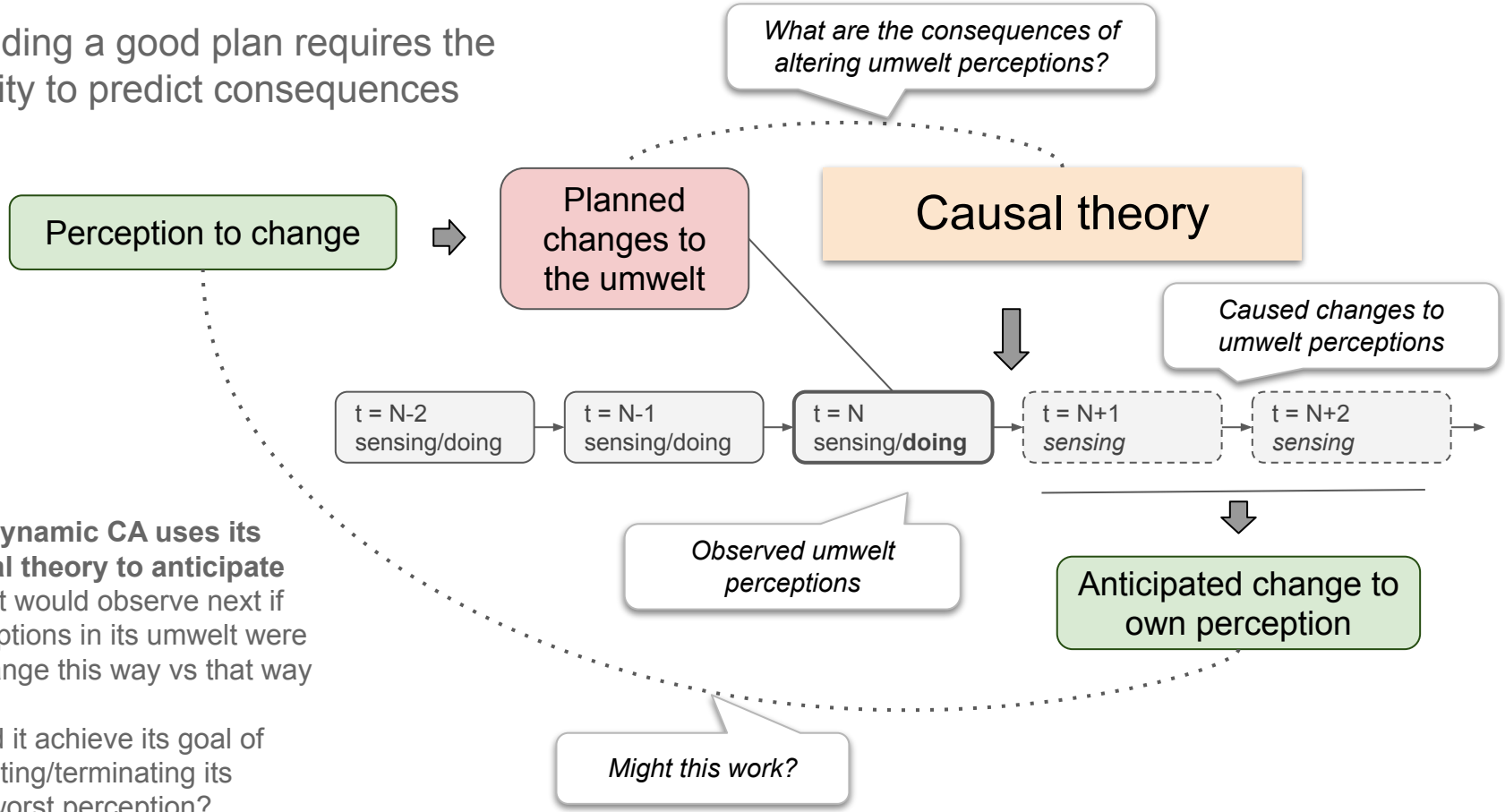
Each individual directive is a delegated, **prioritized goal** to initiate, persist or terminate a contributing umwelt perception

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Building a good plan requires the ability to predict consequences



The dynamic CA uses its causal theory to anticipate what it would observe next if perceptions in its umwelt were to change this way vs that way

Would it achieve its goal of persisting/terminating its best/worst perception?

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When a dynamic CA receives directives (goals to accomplish) from a parent, it decides which ones, if any, it can accomplish and whether they supercede its own self-assigned goals

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A CA does not micromanage its umwelt

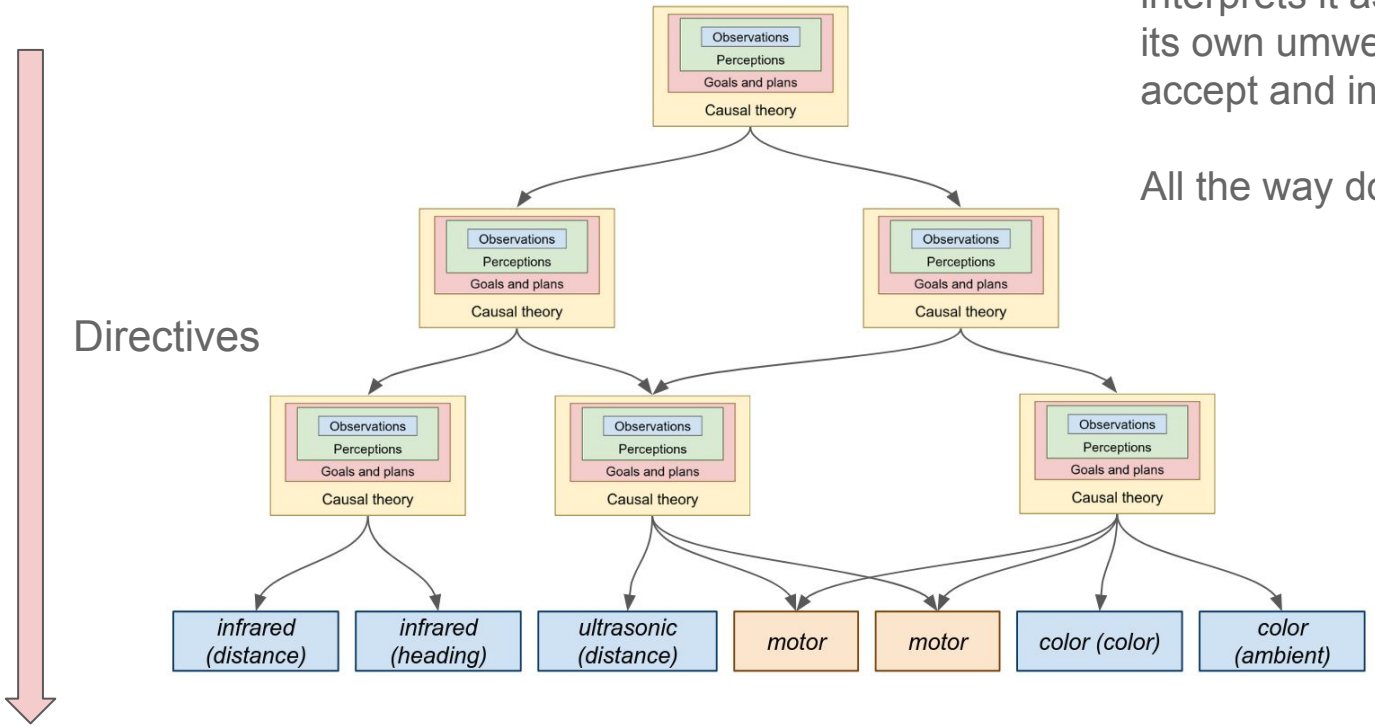
Umwelt CAs *interpret* the directives they receive and accept by making their own plans

The collective of CAs is a multi-scale competency

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For each goal it takes on, a CA interprets it as directives it sends to its own umwelt CAs, for them to accept and interpret, and so on

All the way down to effector CAs



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A dynamic CA keeps track of plans sent and accepted, and the goals they were meant to achieve

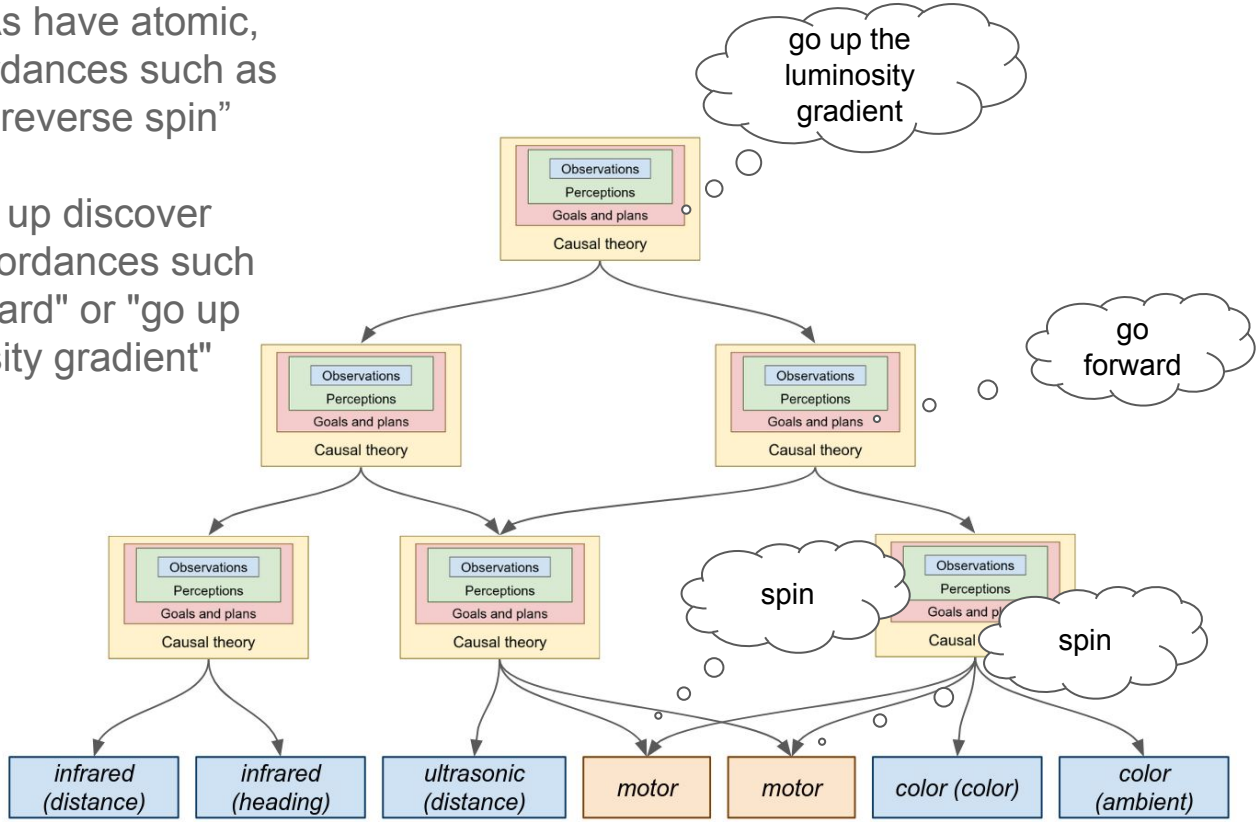
Plans that appear a posteriori to have achieved their goal are remembered as **affordances** for later reuse

Affordances become *habits*

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Effector CAs have atomic, preset affordances such as “spin” and “reverse spin”

CAs higher up discover abstract affordances such as "go forward" or "go up the luminosity gradient"



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Affordances are discovered and named by dynamic CAs, not by the programmer

They need not make sense to an external observer

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Causal theories as logic programs

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A CA's observes its umwelt one **time frame** after another

Observations made within a time frame are considered synchronous

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The more abstract the CA, the longer its time frame for observing, perceiving and acting

A more abstract CA has a bigger “cognitive light cone”

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A dynamic CA constructs a **causal theory** to make sense of what it observes within time frames

It is a form of **induction**, that is, the production of general rules from specific examples

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A causal theory is a set of **if-then rules** about

- what observations ought to co-occur in a time frame
- what observations can not
- what observations follow previous observations

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For example

- *If the distance is getting shorter then nothing is being touched*
- *The distance can not be getting both smaller and larger at once*
- *If both wheels spin forward then the distance will get smaller*

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A causal theory can posit hidden (latent) facts needed to make sense of actual observations

For example, an unobserved obstacle could be imagined by the causal theory to explain an inability to move

This is a form of **abduction**, that is, the inference of likely causes from observed effects given general rules

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The rules making up a causal theory are expressed as a **logic program** (a collection of deductive rules)

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The logic program embodying a causal theory is *discovered* from a small amount of observations, without human intervention

There is no pre-training

Causal modeling is continuous over a CA's lifetime

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A dynamic CA runs its logic program to predict the next observations, and to do what-if reasoning before acting

The logic program serves as a *generative model*

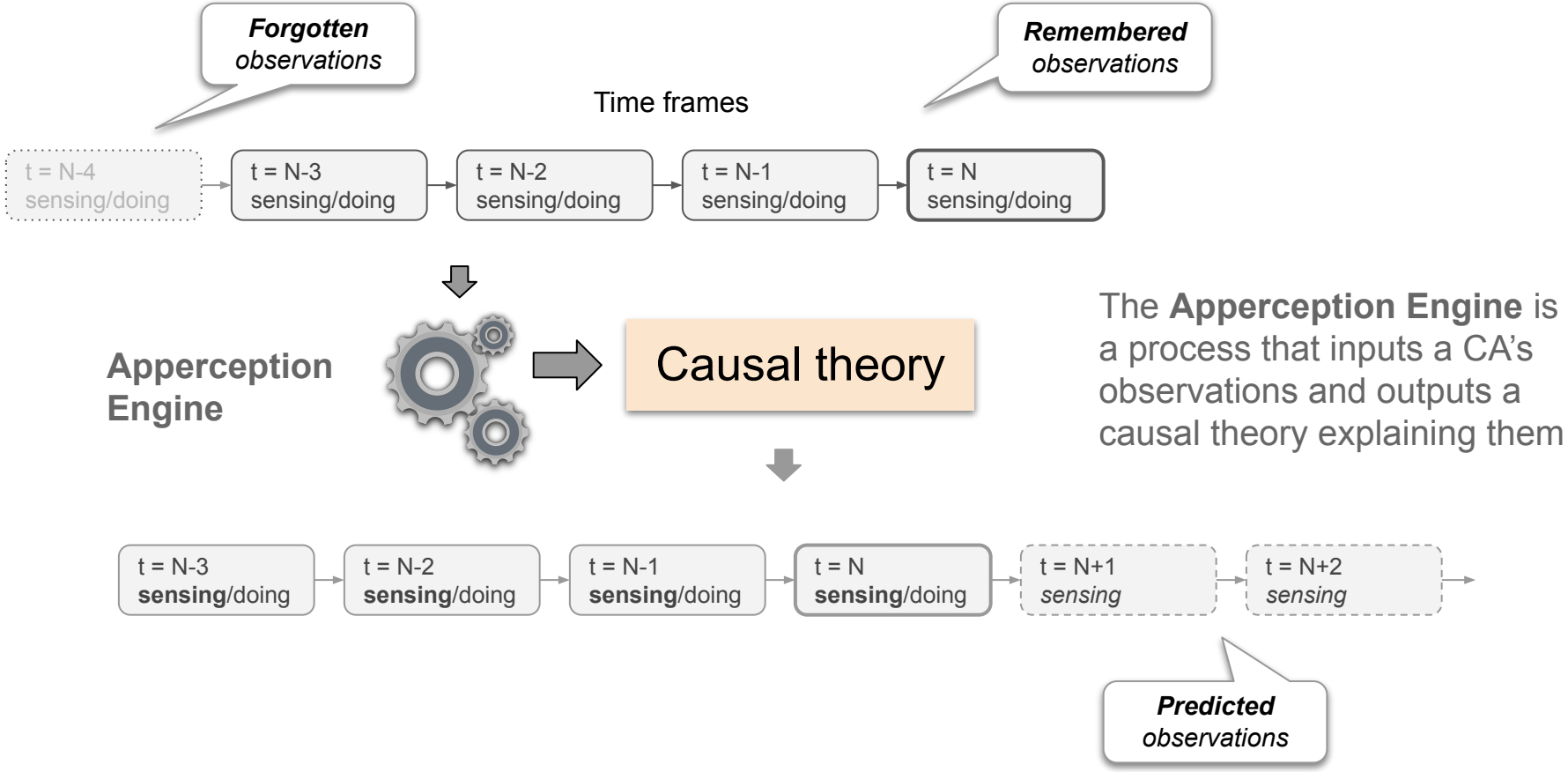
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The Apperception Engine

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A dynamic CA requests a logic program (encoding a causal theory) from the **Apperception Engine** by providing it with its latest observations

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The **Apperception Engine** is a process that inputs a CA's observations and outputs a causal theory explaining them

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Dynamic CAs call upon the Apperception Engine when they experience too many surprises using their current causal theories

Or when they fail consistently to predict the consequences of acting upon their umwelts

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Finding, from remembered observations, the *simplest* logic program that makes *accurate* predictions about future observations, is a hard *search problem*

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The space of all candidate logic programs is extremely large

Finding an accurate causal theory could take too long for it to be of use

We need to relax the conditions for success and look for a *good enough* program (simple enough and accurate enough)

We also need to bias the search so it productively ignores most candidate programs

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Kant's **Synthetic Unity of Apperception** specifies what makes a causal theory coherent (well-formed) and what gives it explanatory power

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It provides strong *inductive biases* that exclude much of the search space

An Apperception Engine is a program that applies Kant's Synthetic Unity of Apperception to find logic programs that encode useful and meaningful causal theories

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The original Apperception Engine was conceived and built by *Richard Evans et al.* of Google DeepMind

I re-implemented and optimized it so that good enough causal theories have a good chance to be found in “useful time”

See "Finding causal theories quickly enough - building a responsive Apperception Engine" at https://zenodo.org/records/15512255

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The Karma System

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The project's public name is **Karma**

The design notes and code are in the **karma_system** repo and sub-repos

https://github.com/jfcloutier/karma_system

Everything is open-sourced

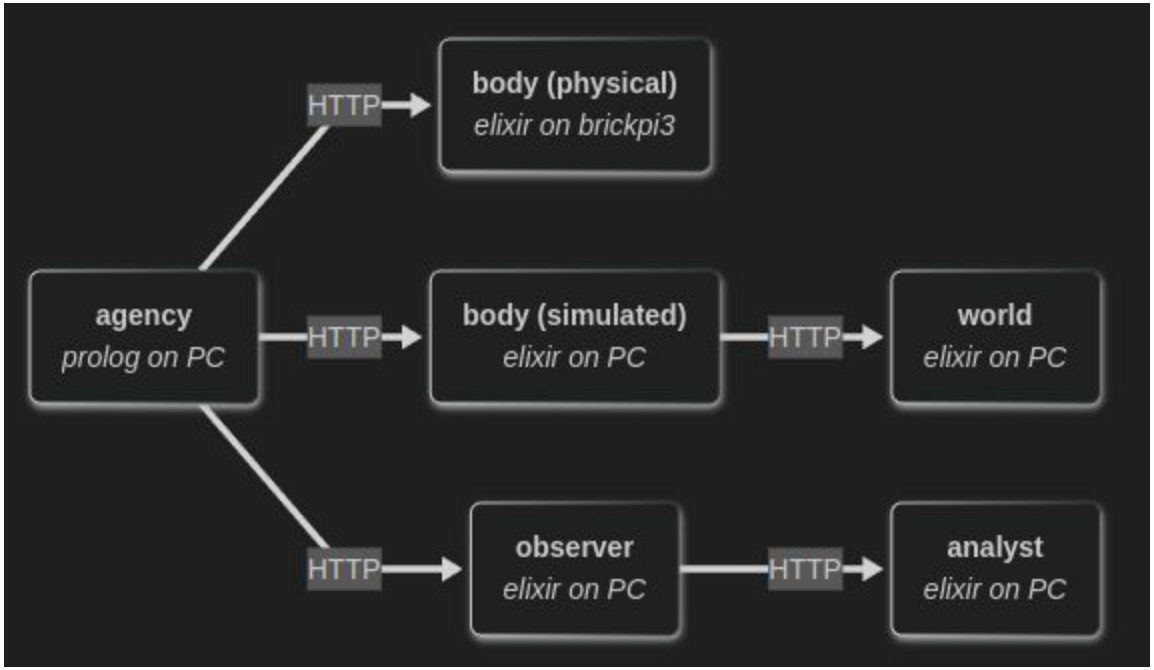
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The Karma sub-repos

- **karma_actors** - Actor Model framework in Prolog
- **karma_apperception** - A practical Apperception Engine
- **karma_agency** - Prolog code implementing agency using actors
- **karma_prolog_utils** - Prolog utility modules
- **karma_body** - Elixir code giving access to the robot's sensors and motors, actual or simulated
- **karma_world** - Elixir code implementing a virtual environment
- **karma_observer** - Code providing an observer view on the robot's cognition
- **karma_analyst** - Code facilitating the FEP analysis of data gathered from running the robot

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The Karma subsystems communicate over a local network



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Progress report (as of November 2025)

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Done - subject to future iterations

- Conceptual and implementation design (karma_system)
- Interface to the robot's sensors and effectors (karma_body)
- Virtual grid world (karma_world)
- Apperception Engine (karma_apperception)
- Runtime platform for CAs (karma_actors)
- Sensor and effector CAs (karma_agency)

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In progress

- Dynamic CAs (karma_agency)

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To do

- Physical recreation of the grid world
- Observer subsystem
- Analyst subsystem
- Experimental runs and data collection
- FEP data analysis

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Parting thought

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Should I succeed at programming a robot with felt experiences, who makes sense of its world, and strives to survive

Could there be something it is like to be this robot?

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jf.cloutier@activeinference.institute

#sym_cog_robotics on the Active Inference Institute's Discord

https://github.com/jfcloutier/karma_system

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